Overreaction in Household Inflation Expectations: Evidence from China

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Abstract

This paper asks whether households overreact to salient relative-price shocks when forming inflation expectations. I combine province-level meat CPI shocks with four waves of the China Family Panel Studies (2016–2022) and estimate a forecast-error design with region-by-wave fixed effects. Within the same region and wave, provinces with larger meat-price increases have substantially more negative subsequent forecast errors: $\hat{\beta}_3 = -6.07$ (wild cluster bootstrap $p < 0.001$, 31 clusters), implying about -1.70 percentage points for a one-standard-deviation shock. A signal-extraction benchmark implies a rational bound $|\beta_3^R| \leq 0.05$ under any persistence belief, so the estimated response is far too large to be reconciled with rational updating alone. The mechanism is a wedge decomposition: meat shocks raise expectations but do not raise forward headline CPI because the ASF-driven price spike later reverses. Placebo tests show that grain shocks with comparable CPI weight do not generate the same pattern. The evidence supports salience-driven diagnostic overreaction in household inflation beliefs.

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1 Introduction

Households exposed to large pork-price shocks systematically overpredict future inflation. Using province-level meat CPI variation from the National Bureau of Statistics matched to four waves of the China Family Panel Studies (2016–2022), I estimate that a one-unit increase in the cumulative meat log-price shock generates a forecast error of -6.07 percentage points under the preferred specification with region-by-wave fixed effects (wild cluster bootstrap $p < 0.001$, 31 province clusters); a one-standard-deviation shock (0.28 log points) implies a forecast error of -1.70 percentage points. The negative sign means that households in high-shock provinces expected more inflation than materialised. A signal-extraction model calibrated to the actual persistence of meat-price shocks bounds the rational forecast error at $|\beta_3^R| \leq \omega \approx 0.05$ per unit of the shock. The estimated coefficient exceeds this bound by a wide margin, and no combination of persistence assumptions within the plausible range closes the gap. The excess is consistent with diagnostic overreaction (Bordalo, Gennaioli, and Shleifer; Bordalo et al.).

Why does this matter for macroeconomics? Inflation expectations anchor wage bargaining, consumption timing, and portfolio allocation. If a large share of the population systematically overpredicts inflation in response to salient relative-price movements, the expectation wedge feeds back into real activity through precautionary saving and consumption compression. In China, where pork accounts for roughly 60 percent of household meat consumption and food carries an official CPI weight near 28 percent, the scope for salience-driven belief distortion is larger than in economies with lower food shares or more diversified protein markets. The 2018–2019 African Swine Fever (ASF) episode, which reduced the national hog herd by an estimated 30–55 percent, provides sharp cross-province variation in the size of the salience signal.

The identification rests on a three-equation system that separates the belief channel from the price-level channel. The primary test is a reduced-form regression of household forecast errors on the province-level meat-price shock (Equation 3). Two decomposition equations ask whether the shock raises household expectations (Equation 1) and whether it predicts forward headline CPI (Equation 2). The expectation response is positive across specifications but imprecisely estimated under the demanding region-by-wave fixed-effect structure; the forecast-error result is the more powerful test because it measures the joint outcome of expectations minus realised inflation directly and does not require the expectation channel to be individually precise. The treatment is measured by the NBS and merged from outside the CFPS survey, so the household's current expectation never appears on both sides of any equation.

I make three contributions. First, I provide a forecast-error test of household inflation overreaction that avoids the mechanical negative correlation in standard revision-error designs. The identifying variation comes from an external, supply-driven price signal, not from within-survey belief revisions. Second, I derive a calibrated rational benchmark from a signal-extraction model and show that the estimated forecast error exceeds the rational bound under any persistence assumption. This separates behavioural overreaction from rational forecast errors due to transitory-shock uncertainty. Third, I show that the overreaction is commodity-specific. Meat shocks generate forecast errors while grain shocks of comparable CPI weight do not, consistent with salience-driven rather than rational updating.

The paper connects to three literatures. Diagnostic expectations (Bordalo, Gennaioli, and Shleifer; Bordalo et al.) predict that agents overweight signals made representative by recent experience; the forecast-error pattern I document is a direct counterpart of this prediction. Rational inattention (Sims; Maćkowiak and Wiederholt) predicts selective attention to informative signals; the commodity asymmetry is consistent with disproportionate attention to high-frequency prices. Experience-based learning (Malmendier and Nagel; Kuchler and Zafar) predict that personal purchase experience shapes expectations more than official statistics; the meat channel operates through precisely these signals.

Three limits apply. The CFPS is biennial, so I cannot trace within-year adjustment dynamics. Province-level meat CPI conflates supply and demand movements; ASF provides a plausible supply instrument, but the treatment is an imperfect proxy for the supply component alone. With 31 province clusters, inference throughout relies on the wild cluster bootstrap.

The remainder of the paper proceeds as follows. Section 2 provides institutional background. Section 3 develops the conceptual framework. Section 4 describes the data. Section 5 presents the empirical strategy. Section 6 reports main results. Section 7 presents commodity placebo evidence. Section 8 contains robustness checks. Section 9 discusses alternative interpretations and policy implications. Section 10 concludes.

2 Institutional Background and Related Literature

2.1 Food Prices in Chinese Inflation

Food is the single largest category in the Chinese CPI basket. The National Bureau of Statistics weights food and non-alcoholic beverages at approximately 28 percent of the overall index, roughly twice the comparable share in high-income economies. This weight reflects both the expenditure share of food in Chinese household budgets and the NBS's plutocratic aggregation method. Because food expenditure shares decline with income (Engel's law), the effective food weight has been falling over time—from roughly 34 percent in the early 2000s to 28 percent after the 2016 basket revision—but remains substantially higher than in OECD economies.

The NBS constructs the CPI using a modified Laspeyres index with a five-year base-period update cycle. A consequential change occurred in 2016, when the NBS merged the previously separate urban and rural CPI baskets into a unified national basket. Before 2016, rural and urban price indices used different item lists and different outlet samples, creating discontinuities in province-level CPI series at the merge point. I begin the analysis in 2016 to work with a consistent basket throughout.

Within food, pork alone accounts for around 60 percent of total meat consumption in China. The NBS does not publish the exact weight of individual food items, but researchers who have backed out pork's contribution by regressing headline CPI on sub-indices consistently place the pork weight at 2.5–4.5 percent for the 2016–2020 basket, with higher effective weights in provinces where per-capita pork consumption exceeds the national mean.

That official weight understates pork's effective salience. Pork is purchased frequently: unlike durable categories whose prices households observe intermittently, pork is a daily staple for a large share of the population. Purchase frequency amplifies the informational weight that consumers assign to price changes relative to budget share (Cavallo, Cruces, and Perez-Truglia). The same logic applies with particular force in China, where wet markets are the primary retail channel for fresh protein and where household members observe pork prices at every market visit—often daily in urban areas.

Historically, pork price cycles have been the single most important contributor to headline CPI volatility in China. The so-called "pig cycle" has generated four distinct peaks in pork retail prices since 2005: in 2007–2008, 2010–2011, 2015–2016, and the ASF-driven peak of 2019–2020. During each episode, year-on-year pork price inflation exceeded 50 percent for at least two consecutive months, and the pork sub-index accounted for 60–80 percent of the total variation in food CPI. The 2019 episode was the most extreme in the modern CPI record: the NBS meat and poultry sub-index reached 122 percent year-on-year in February 2020.

The baseline 28 percent food weight creates a strong prior that food-price movements are informative about aggregate inflation. When pork is expensive, households that weight recent experience heavily form upward-revised expectations even if non-food prices are stable. This channel is not irrational in a narrow accounting sense because food prices enter headline CPI. The issue is magnitude. The expectation response can exceed what food weights justify.

2.2 African Swine Fever as a Natural Experiment

African Swine Fever (ASF) is a highly contagious viral disease affecting domestic and wild pigs with near-100 percent mortality in infected herds. It has no licensed vaccine, no cure, and no transmission pathway to humans. These features make ASF exceptionally difficult to contain once it enters a country's swine population.

China confirmed its first domestic case in Shenyang, Liaoning province, in August 2018. Epidemiological tracing suggested that contaminated swill feed and long-distance live-pig transport were the initial vectors. Within two months, ASF had been confirmed in Heilongjiang, Henan, Jiangsu, and Zhejiang, spanning a 2,000-kilometer north-south corridor. By February 2019, every mainland province except Hainan had reported at least one outbreak.

The geographic pattern of spread reflected China's live-pig trade network. North-eastern provinces (Liaoning, Heilongjiang, Jilin) are net exporters of pigs to consuming regions in the southeast (Guangdong, Fujian, Zhejiang). Long-distance truck transport moved infected animals across provincial borders before local outbreaks were detected.

The Ministry of Agriculture imposed interprovincial transport bans in September 2018, but enforcement was incomplete, and small-scale farmers—roughly 45 percent of national pork production—continued informal cross-border sales.

Between August 2018 and mid-2019, the official culling count exceeded 1.1 million pigs, but independent estimates from the FAO placed the total herd reduction at 30 to 55 percent of the pre-ASF national stock—roughly 120 to 210 million animals. Wholesale pork prices remained roughly flat through the first five months after the initial outbreak as frozen stocks buffered supply. Starting in March 2019, prices began a sustained increase. By October 2019, the NBS retail pork price index stood at approximately double its August 2018 level. The price increase was geographically uneven: provinces with larger pre-ASF herd densities (Sichuan, Henan, Hunan, Hubei) experienced the sharpest increases, while coastal provinces with cold-chain import infrastructure partially offset domestic supply losses with imports from the EU, Brazil, and the US.

The price shock reversed between 2020 and 2022. As surviving large-scale producers expanded capacity, the national herd rebuilt faster than expected; by mid-2021, hog inventories had returned to 95 percent of pre-ASF levels. Pork prices fell below pre-ASF levels in late 2021. This reversal is consequential for identification. Households who expected continued high inflation after observing the 2018–2020 meat-price surge were wrong about the direction, not just the magnitude, of subsequent price movements.

Two features of ASF matter for the identification strategy. First, the disease spread through biological transmission in pig populations, not as a consequence of local economic conditions or household demand. Cross-province variation in meat prices is therefore plausibly orthogonal to demand shocks that might independently affect inflation expectations. Second, the province-level treatment intensity was determined by pre-existing herd geography—centuries of agricultural specialisation, climate suitability, and feed-cost differentials—rather than by contemporary household characteristics. This pre-determination rules out the concern that provinces with systematically different expectation-formation processes selected into larger shocks.

2.3 Related Literature

Related evidence falls into three strands.

2.3.1 Overreaction in Expectations

Coibion and Gorodnichenko provide a foundational test using the information-rigidity framework. Under noisy information or sticky updating, forecast revisions should positively predict subsequent forecast errors. A negative revision-error slope, conversely, indicates overreaction. Coibion and Gorodnichenko find that US professional forecasters display positive revision-error covariance for most macroeconomic variables, consistent with information rigidity rather than overreaction.

Bordalo et al. revisit this test with a longer panel of professional forecasts across six macroeconomic variables. They document a sign switch: at short horizons, forecast revisions negatively predict errors (overreaction), while at longer horizons the relationship is positive (underreaction). The diagnostic expectations model of Bordalo, Gennaioli, and Shleifer generates exactly this horizon-dependent pattern through a single parameter θ that governs the degree to which agents overweight the likelihood ratio of the signal.

Angeletos, Huo, and Sastry develop a model in which dispersed information generates aggregate patterns observationally similar to overreaction even when individual agents update rationally. The province-level design here partially addresses this concern: because the treatment varies across provinces within a given wave, the identification does not rely on aggregate time-series patterns that could be confounded by general equilibrium feedback.

I provide a household-level test that exploits an external commodity shock rather than within-survey revisions. The meat-shock design avoids the mechanical correlation that arises in the standard revision-error framework: the treatment is an NBS price statistic that does not contain any household expectation data, and the dependent variable (forecast error) is constructed from the household's expectation and subsequent realised CPI—both measured independently of the treatment.

2.3.2 Salience and Inflation Expectations

Cavallo, Cruces, and Perez-Truglia show that US households weight frequently purchased goods more heavily in their inflation perceptions than the official CPI weights imply. Respondents systematically overweight price changes for items they buy weekly (groceries, gasoline) relative to items they buy annually (appliances, insurance). The overweighting is large: perceived inflation tracks a frequency-weighted index roughly twice as closely as it tracks the official CPI.

D'Acunto et al. extend this logic to European households using the European Commission's Consumer Survey. Country-level variation in the prices of frequently purchased goods predicts cross-country variation in household inflation expectations after controlling for official CPI. The effect is asymmetric: price increases in salient categories raise expectations by more than price decreases lower them.

Malmendier and Nagel document that personally experienced inflation has a stronger influence on expectations than historical data alone. Using the Michigan Survey of Consumers, they show that respondents who lived through periods of high inflation report systematically higher expectations decades later. Kuchler and Zafar document this pattern for housing expectations, showing that personal experience with local house-price appreciation predicts national housing return forecasts.

These findings motivate the commodity-specific design used here. If salience operates through purchase frequency and personal exposure, meat and grain shocks should produce different expectation responses despite similar CPI weights. The placebo test uses this asymmetry: grain shocks of comparable CPI weight do not generate forecast errors, which rules out mechanical pass-through as the explanation for the meat-shock result.

2.3.3 China-Specific Evidence

Household inflation expectations in China have received limited attention relative to the US and European literatures. The primary data source is the People's Bank of China (PBoC) Urban Depositor Survey, a quarterly survey of roughly 20,000 depositors in 50 cities that uses an ordinal scale to elicit expected price changes.

Chen and Zha document how financial-policy regimes shape China's macroeconomic dynamics. Direct evidence on household inflation-expectation formation in China remains limited and is often based on aggregate diffusion indices from the PBoC Urban Depositor Survey. The aggregate nature of these data limits identification of cross-sectional expectation responses, and the survey samples depositors rather than the general population.

The CFPS micro data used here address both limitations. The CFPS is a nationally representative panel covering both urban and rural households across 25 provinces.

Province-level treatment variation allows me to test whether the expectation response differs across locations with different shock magnitudes, and the panel dimension allows me to control for household fixed effects.

The ASF episode provides identifying variation that has not been exploited in prior work on Chinese inflation expectations. Existing studies of ASF in the economics literature focus on supply-chain disruption, international trade effects, and food security, but none examines the household-level belief response. This analysis fills that gap by connecting province-level price consequences of ASF to the inflation expectations of households exposed to those prices.

3 Conceptual Framework

This section derives a rational benchmark for the household forecast error and shows how diagnostic expectations generate overshooting. The model is deliberately parsimonious: its purpose is to discipline the empirical test, not to estimate structural parameters. Full derivations appear in Appendix A.

3.1 Setup

A household in province p at survey wave t observes the cumulative meat-price change s_{pt} between waves $t-1$ and t. The household must forecast headline CPI inflation $\pi_{p,\,t\to t+1}$ over the subsequent inter-wave period.

Meat enters headline CPI with weight $\omega \approx 0.05$. The meat-price shock has persistence $\rho \in [0,1]$: a fraction ρ of the current shock carries forward into the next period's headline inflation. Formally, the headline CPI response to a meat shock of size s is

$$\mathbb{E}[\pi_{p,\,t\to t+1} \mid s_{pt}] = \bar{\pi} + \omega\,\rho\,s_{pt}, \tag{1}$$

where $\bar{\pi}$ is unconditional expected inflation. The household does not know ρ with certainty but holds a prior belief $\hat{\rho}$.

I focus on persistence uncertainty rather than weight uncertainty because CPI weights are approximately stable across the sample period, so persistent disagreement about ω

would produce a constant expectation bias absorbed by fixed effects. Persistence uncertainty, by contrast, interacts with the size of the shock and generates a coefficient on s_{pt} in the forecast-error regression.

3.2 Rational Benchmark

Under rational expectations, the household forecasts $\mathbb{E}_t^R[\pi_{p,t\to t+1}] = \bar{\pi} + \omega\,\hat{\rho}\,s_{pt}$, where $\hat{\rho}$ is the household's best estimate of persistence. The forecast error is

$$\mathrm{FE}_{pt}^R = \pi_{p,t\to t+1} - \mathbb{E}_t^R[\pi_{p,t\to t+1}] = \omega(\rho - \hat{\rho})\,s_{pt} + \eta_{pt}, \tag{2}$$

where η_{pt} collects shocks to headline CPI that are orthogonal to the meat signal. If the household overestimates persistence ($\hat{\rho} > \rho$), it expects more meat-driven inflation than actually occurs, and the forecast error is negative in provinces with positive shocks. If the household underestimates persistence, the forecast error is positive in high-shock provinces.

The coefficient on s_{pt} in a regression of the forecast error on the meat shock is therefore bounded:

$$\beta_3^R = \omega(\rho - \hat{\rho}). \tag{3}$$

Even in the worst case—where households believe the shock is permanent ($\hat{\rho} = 1$) but it is fully transitory ($\rho = 0$)—the rational forecast-error coefficient is $\beta_3^R = -\omega \approx -0.05$. Under any intermediate belief, $|\beta_3^R| < \omega$. The rational benchmark therefore predicts $|\beta_3| \leq 0.05$ per unit of the meat shock.

This bound is the central object of the empirical test. The number 0.05 is not a free parameter—it is the CPI weight on meat, determined by the NBS basket construction. Any rational household, regardless of its belief about persistence, generates a forecast-error coefficient no larger than the mechanical weight of meat in the index, because persistence governs the *duration* of the meat contribution, not its *size per period*.

3.3 Diagnostic Extension

Under diagnostic expectations (Bordalo, Gennaioli, and Shleifer), the household overweights the state that recent signals make more representative. A large positive meat-price shock makes high-inflation states more salient, so the household inflates its forecast by a factor $\theta > 0$ beyond the rational update:

$$\mathbb{E}_t^{\theta}[\pi_{p,\,t\to t+1}] = \bar{\pi} + \omega\,\hat{\rho}\,s_{pt} + \theta\,\omega\,s_{pt}. \tag{4}$$

The additional term $\theta\,\omega\,s_{pt}$ arises because the household tilts its subjective probability distribution toward states with high likelihood ratios. When $\theta = 0$, the household reverts to the rational benchmark; when $\theta > 0$, it places too much probability mass on high-inflation outcomes after observing a positive meat-price shock.

The diagnostic forecast error is

$$\mathrm{FE}_{pt}^{\theta} = \omega(\rho - \hat{\rho})\,s_{pt} - \theta\,\omega\,s_{pt} + \eta_{pt}, \tag{5}$$

so the regression coefficient becomes $\beta_3^{\theta} = \omega(\rho - \hat{\rho} - \theta)$. With $\theta > 0$, the forecast error is more negative than under rationality for any belief $\hat{\rho}$ and any true persistence ρ.

The forecast-error regression is a sharper test than the expectation regression. The expectation coefficient β_1 depends on $\omega(\hat{\rho} + \theta)$, so distinguishing the rational component from the diagnostic component requires an independent estimate of $\hat{\rho}$, which is not directly observed. The forecast-error coefficient β_3, by contrast, is bounded by ω under rationality regardless of $\hat{\rho}$. Any estimate of $|\hat{\beta}_3|$ exceeding 0.05 rejects rationality without requiring knowledge of the household's belief about persistence.

3.4 Testable Implications

The framework generates three predictions that the data can discriminate.

Prediction 1: Bounded forecast errors under rationality. Under rationality, the forecast-error coefficient satisfies $|\beta_3| \leq \omega \approx 0.05$. If the estimated $\hat{\beta}_3$ exceeds this bound in absolute value, rationality is rejected regardless of the household's belief about persistence. The bound is conservative: it assumes the worst-case belief ($\hat{\rho} = 1$ when

$\rho = 0$); under any more accurate belief, the rational coefficient is smaller. In the data, the estimated coefficient is $\hat{\beta}_3 = -6.07$, which exceeds the rational bound by a factor of 120.

Prediction 2: Excess expectation response. Under diagnostic expectations the expectation response exceeds the rational update: $\hat{\beta}_1 > \omega\,\hat{\rho}$. The expectation result complements the forecast-error test by showing *where* the excess forecast error comes from: through the numerator (expectations are too high) rather than the denominator (realised CPI is too low). In the preferred specification, $\hat{\beta}_1 = 0.242$ (s.e. $= 0.174$), which is positive but imprecisely estimated due to the ordinal nature of the expectation variable.

Prediction 3: Commodity asymmetry. A household that overweights salient signals should respond more to meat shocks (high purchase frequency, high media coverage) than to grain shocks of comparable CPI weight. The diagnostic model predicts that θ is larger for meat than for grain because the purchase-frequency channel amplifies the representativeness of meat-price signals. Grain prices in China are stabilised by government procurement and strategic reserves, so they vary less and attract less household attention. The placebo test in Section 7 confirms this asymmetry: grain shocks produce near-zero forecast errors.

3.5 Mapping to Empirical Equations

The three predictions map directly to the three-equation system estimated in Section 5. Equation 1 regresses household expectations on the meat shock and tests whether $\hat{\beta}_1 > 0$. Equation 2 regresses forward realised headline CPI on the meat shock and estimates the actual pass-through $\omega\,\rho$. Equation 3 regresses the forecast error (realised CPI minus expectation) on the meat shock and tests whether $|\hat{\beta}_3| > \omega$.

The empirical strategy uses province-by-wave fixed effects in the preferred specification. In the model, these fixed effects absorb $\bar{\pi}$ and any province-by-wave shocks common to all households in a given cell. The identifying variation is the deviation of a province's meat shock from the region-by-wave mean. The model predicts that the forecast-error bound $|\beta_3| \leq 0.05$ holds even in this demanding specification because the bound derives from the CPI weight, not from the level of the shock.

Section 6.4 compares the estimated $\hat{\beta}_3$ against the rational bound calibrated to the actual persistence of meat-price shocks in the data.

4 Data

4.1 China Family Panel Studies

The China Family Panel Studies (CFPS) is a nationally representative biennial household panel survey administered by the Institute of Social Science Survey (ISSS) at Peking University (Institute of Social Science Survey, Peking University). The baseline wave in 2010 drew a stratified, multi-stage probability sample of approximately 16,000 households in 25 mainland provinces, covering roughly 95 percent of the national population. The survey has been fielded in 2010, 2012, 2014, 2016, 2018, 2020, and 2022. I use the four waves from 2016 to 2022, which overlap with the availability of province-level meat CPI data from the NBS. For auxiliary cross-sectional checks in companion specifications, I draw CHFS source files from the China Household Finance Survey and Research Center at Southwestern University of Finance and Economics (China Household Finance Survey and Research Center, Southwestern University of Finance and Economics).

Response rates are high by international panel standards. The baseline household response rate was approximately 81 percent, with re-contact rates between 75 and 85 percent in subsequent waves. Attrition is non-random—younger, more educated, and urban-resident households are more likely to exit—and I include age, education, and urban indicators as controls where available.

The inflation expectation variable is a three-point ordinal question (item qp201) that asks respondents: "Compared with the current period, do you think the price level in the coming year will rise, stay the same, or fall?" I code the response as $\mu_{it} \in \{-1, 0, 1\}$, where 1 indicates expected price increases and -1 indicates expected decreases. The question wording refers to "price level" (物价水平) rather than the CPI specifically, which is consistent with the salience channel: if respondents map "price level" to the prices of goods they buy frequently, their responses will be disproportionately influenced by salient price changes.

The coarseness of the three-point scale attenuates the estimated coefficients: respondents who revise their beliefs upward by different amounts all register as $+1$. This attenuation bias works against finding overreaction, so any detected overreaction is a lower bound.

The estimation sample contains 90,133 household-wave observations across 31 provinces and four waves. The forecast-error sample is smaller (69,533 observations) because the 2022 wave lacks forward realised headline CPI at the time of data construction. Demographic controls—age, a high-education indicator, and an urban-residence indicator—are available for a subsample of approximately 22,100 observations.

4.2 Province-Level CPI Data

The National Bureau of Statistics (NBS) publishes monthly consumer price indices at the province level for major sub-categories. I use the meat and poultry sub-index, the grain sub-index, and the headline CPI index. Province-level meat CPI is available from January 2016 onward; grain and headline CPI are available from January 2012.

Geographic coverage spans all 31 mainland provinces. Two potential measurement concerns arise. First, the outlet sample may not fully represent rural areas, where wet markets are the dominant channel for fresh protein. If rural meat prices move differently from the sampled outlets, the NBS meat sub-index may understate the price variation experienced by rural CFPS households, attenuating the estimated coefficients toward zero. Second, the transition from separate urban and rural CPI baskets to a unified basket in 2016 may create a level shift in some province-level series. I avoid this problem by beginning the analysis in 2016 and using cumulative log changes, which difference out level shifts.

The NBS reports these sub-indices in $\textit{previous month} = 100$ form: a value of 105 means the price level rose 5 percent relative to the preceding month. I convert each monthly observation to a log change by dividing by 100 and taking the natural logarithm; because the base is the previous month, this yields the exact month-on-month log growth rate. I then sum these monthly log changes over the inter-wave period to obtain the cumulative treatment.

4.3 Treatment Construction

The treatment variable is the cumulative log change in provincial meat CPI between two consecutive CFPS waves:

$$\text{MeatShock}_{p,\,t-1\to t} = \sum_{m\in\mathcal{M}(t-1,t)} \log\left(\frac{\text{MeatCPI}_{pm}}{100}\right), \tag{6}$$

where MeatCPI_{pm} is the NBS meat and poultry sub-index for province p in month m, and $\mathcal{M}(t-1,t)$ is the set of months between the midpoints of two consecutive CFPS fieldwork periods. The grain shock is constructed identically from the grain sub-index.

I define the inter-wave period using the midpoint of each wave's fieldwork window. The 2016 wave was fielded primarily between June and December 2016; the 2018 wave between June and November 2018; the 2020 wave between July 2020 and January 2021 (delayed by COVID-19); and the 2022 wave between June and December 2022. This midpoint convention ensures that the treatment captures the price changes that the typical respondent experienced between consecutive interview dates.

The identifying variation operates at the province×wave level. The sample contains 93 province×wave cells across 31 provinces and three to four waves, depending on forward CPI availability. All standard errors are clustered at the province level (31 clusters), and inference uses the wild cluster bootstrap to account for the small number of clusters.

4.4 Descriptive Statistics

Table 1 reports summary statistics for the estimation sample.

The mean inflation expectation is -0.32 on the $[-1, 1]$ scale, indicating that the majority of respondents expect prices to stay the same or fall. This finding is consistent with two features of the data. First, the sample period includes the post-ASF price reversal of 2021–2022, during which headline CPI inflation fell below 1 percent. Second, the $\{-1, 0, 1\}$ coding treats "stay the same" as the midpoint.

The mean forecast error is 4.67 percentage points. The positive sign reflects the mechanical asymmetry of the ordinal coding: when respondents report expectations of zero or below but realised CPI inflation is positive, the forecast error is positive. The

Table 1: Descriptive Statistics

Variable	N	Mean	SD	Min	Median	Max
Inflation expectation (μ_{it})	90,133	-0.32	0.81	-1.00	-1.00	1.00
Forecast error (FE_{it})	69,533	4.67	1.88	-0.21	5.07	8.14
Meat shock (s_{pt})	90,133	0.05	0.28	-0.40	-0.01	0.57
Grain shock	90,133	0.04	0.02	-0.07	0.03	0.12
Realized headline CPI (%, ann.)	69,533	4.03	0.96	1.79	3.86	6.14
Age	22,320	40.19	14.27	18.00	39.00	70.00
High education (indicator)	90,133	0.30	0.46	0.00	0.00	1.00
Urban (indicator)	87,414	0.67	0.47	0.00	1.00	1.00

Note. Sample restricted to observations with non-missing province code, inflation expectation, and meat shock.
Inflation expectation coded as $\{-1, 0, 1\}$ (prices fall, stable, rise).
Forecast error = realized headline CPI (%) $-$ expectation scaled to CPI units ($\{-2, 0, 2\}$).
Demographics available for a subsample; N varies across rows.

regression analysis controls for this level effect through fixed effects; the coefficient of interest captures the *differential* forecast error in high-shock versus low-shock provinces, not the mean level.

The mean meat shock is 0.05 log points, with a standard deviation of 0.28. The near-zero mean reflects offsetting waves: the 2018–2020 pair captures the ASF peak (large positive shocks), while the 2020–2022 pair captures the price reversal (large negative shocks). A one-standard-deviation shock of 0.28 log points corresponds to a cumulative meat-price increase of roughly 32 percent ($e^{0.28} - 1 \approx 0.32$) over the inter-wave period.

Forward realised headline CPI averages 4.03 percent (annualised), with moderate cross-province variation driven by the elevated food prices during the 2018–2020 inter-wave period.

5 Empirical Strategy

The strategy exploits province-level meat-price variation as an external salience treatment. The treatment is measured by the National Bureau of Statistics and merged from outside the CFPS survey, so the household's current expectation never appears on both sides of any regression. The primary test is a reduced-form regression of forecast errors on the meat shock. Two decomposition equations identify the channels through which the forecast error arises.

5.1 Treatment Variable

The treatment variable is the cumulative log change in provincial meat CPI between two consecutive CFPS waves, defined in equation (6). Three properties matter for identification. First, it is external to the survey: it contains no information derived from respondents. Second, it is cumulative: a household surveyed in 2020 experienced the full path of meat-price movements since the 2018 wave. Third, it varies at the province level, which allows me to absorb region-by-wave fixed effects and still retain within-region identifying variation.

The cumulative measure is the natural choice given the CFPS survey design. CFPS is fielded every two years, with interviews spanning several months within each wave. Households report expectations once per wave, and those reports reflect whatever price information they accumulated since the prior interview. The cumulative log change between wave midpoints therefore captures the total price signal that a household plausibly processed.

Two measurement considerations deserve attention. First, the NBS meat CPI index combines pork, beef, mutton, and poultry, with pork receiving the dominant weight. During the ASF episode, pork drove the bulk of the meat CPI movement; the index therefore tracks the pork shock with high fidelity during the identification window. Second, the cumulative measure aggregates monthly observations equally. If households weight recent months more heavily, the treatment variable understates effective salience, attenuating the estimated coefficient toward zero.

The 2018–2019 ASF episode provides the sharpest source of variation. Provincial herd losses varied because culling policy, transport restrictions, and initial herd density differed across provinces. This heterogeneity generates cross-sectional dispersion in the treatment that is plausibly orthogonal to province-level demand conditions.

5.2 Three-Equation System

Primary test: forecast errors (Equation 3). The reduced-form implication of overreaction is that households in high-shock provinces subsequently overpredict inflation:

$$\mathrm{FE}_{ipt} = \alpha + \beta_3 \, \mathrm{MeatShock}_{p,\, t-1\to t} + \gamma' X_{ipt} + \delta_{r(p)\times t} + \varepsilon_{ipt}, \tag{7}$$

where $\text{FE}_{ipt} \equiv \pi^{\text{headline}}_{p,\,t\to t+1} - \tilde{\mu}_{ipt}$ is realised provincial headline inflation over the next inter-wave period minus the household's directional expectation scaled to CPI units. A negative forecast error means that the household expected more inflation than materialised. Under overreaction, $\beta_3 < 0$: high-shock provinces generate more negative forecast errors.

The forecast-error equation occupies the primary position because it requires the fewest auxiliary assumptions. The sign of β_3 alone tests overreaction. If households responded rationally, their expectations would adjust by no more than the shock's actual headline CPI weight, and forecast errors would be bounded by $\omega \approx 0.05$ regardless of the household's belief about shock persistence (Section 3). Any $|\hat{\beta}_3|$ exceeding this bound rejects rationality without requiring knowledge of the conversion factor between ordinal expectations and CPI units.

The identification of β_3 comes from within-region variation in meat-price shocks. Two households in different provinces within the same region and survey wave face identical aggregate conditions but different meat-price histories. The coefficient β_3 measures whether the household in the higher-shock province produces a more negative forecast error.

Decomposition: expectations (Equation 1).

$$\mu_{ipt} = \alpha + \beta_1\, \text{MeatShock}_{p,\,t-1\to t} + \gamma' X_{ipt} + \delta_{r(p)\times t} + \varepsilon_{ipt}, \tag{8}$$

where $\mu_{ipt} \in \{-1, 0, 1\}$ is the ordinal inflation expectation. A positive $\hat{\beta}_1$ indicates that higher meat-price shocks shift expectations upward.

Equation 1 identifies whether the shock moves beliefs. If households are inattentive or if meat prices lack salience, β_1 would be zero. Because the dependent variable is ordinal, the coefficient measures shifts in the probability of reporting "up" versus "same" or "down," not a cardinal change in expected inflation. The ordinal scale limits what β_1 can tell us about the magnitude of the expectation response in CPI units, which is precisely why the forecast-error equation is the primary test.

Decomposition: pass-through (Equation 2).

$$\pi^{\text{headline}}_{p,\,t\to t+1} = \alpha + \beta_2\, \text{MeatShock}_{p,\,t-1\to t} + \delta_{r(p)\times t} + u_{pt}, \tag{9}$$

where $\pi^{\text{headline}}_{p,\,t\to t+1}$ is realised province-level headline CPI inflation over the subsequent inter-wave period. This regression runs at the province-wave level (93 observations). If $\beta_2 \leq 0$, the shock does not raise future headline inflation, closing the second half of the overreaction logic.

The pass-through equation pins down what a rational household should have expected. A zero or negative β_2 means the shock reverses. Provinces that received large meat-price increases subsequently saw lower, not higher, headline inflation. The reversal follows ASF supply-shock dynamics. As herds recovered through 2021–2022, pork supply expanded and meat prices fell, dragging down headline CPI in the provinces that experienced the sharpest earlier increases. The 93-observation sample size motivates the wild cluster bootstrap described in Section 5.4.

5.2.1 Decomposition Logic and the Overreaction Test

The three coefficients are linked by an approximate decomposition: $\beta_3 \approx \beta_2 - k\,\beta_1$, where k converts the ordinal expectation into implied inflation units. This identity clarifies the overreaction test.

The forecast error is realised inflation minus expected inflation. Regressing realised inflation on the meat shock gives β_2. Regressing expected inflation on the meat shock gives $k\,\beta_1$. If expectations respond ($\beta_1 > 0$) but realised CPI does not ($\beta_2 \leq 0$), the forecast error is necessarily negative because the household raised its expectation in response to a signal that did not raise the object it was forecasting.

The sign test $\beta_3 < 0$ does not require knowledge of k. The CFPS expectation question is ordinal, so the mapping from $\mu \in \{-1, 0, 1\}$ to CPI units is unknown. But the forecast error is constructed in CPI units, so β_3 is directly interpretable. Whatever k is, if the forecast error is systematically more negative in high-shock provinces, households in those provinces overpredicted inflation.

The decomposition also clarifies why $|\hat{\beta}_3|$ exceeds the rational bound by such a wide margin. Under rational expectations, β_1 should equal $\omega\,\hat{\rho}/k$, where $\omega \approx 0.05$ is meat's CPI weight and $\hat{\rho}$ is the household's persistence belief. Even if $\hat{\rho} = 1$, the rational forecast-error coefficient is bounded by $|\beta_3^R| \leq 0.05$. Finding $|\hat{\beta}_3| \approx 6$ means the actual expectation response far exceeds the rational response.

5.3 Identification Assumptions

The strategy requires two conditions.

The first condition is relevance: the meat-price shock must predict household forecast errors. Pork is widely consumed, purchased frequently, and priced visibly; ASF generated extensive media coverage that was difficult to miss even for inattentive consumers. Three features support relevance: pork accounts for roughly 60 percent of China's meat consumption and is purchased several times per week; the ASF episode received extensive state and social media coverage; and prices at wet markets are posted on visible boards and updated daily.

The second condition is conditional exogeneity: the meat-price shock must be uncorrelated with province-level determinants of forecast errors other than through salience, conditional on fixed effects and controls. Region-by-wave fixed effects absorb regional business cycles, monetary policy transmission, and nationwide shocks hitting all provinces within a region simultaneously. The remaining identifying variation is within-region cross-province differences in meat-price shocks over time.

The main threat is a province-specific demand shock that simultaneously raises meat prices and expectations through a real-income channel. Two features mitigate this concern. First, ASF provides a biologically determined supply shock. The disease entered China through contaminated feed imports and spread along pig transportation corridors, with provincial infection severity determined by initial herd density and proximity to early outbreak sites. Second, the commodity-placebo test in Section 7 shows that grain shocks of comparable magnitude do not produce the same forecast-error pattern. If local demand were driving both prices and expectations, grain prices should respond to the same demand shock and produce a similar expectation effect.

A second concern is that province-level meat-price variation might proxy for local food-price inflation more broadly. Region-by-wave fixed effects partially address this by absorbing region-level food-price trends. The within-region variation in meat prices is largely idiosyncratic to the ASF episode and does not coincide with similar dispersion in non-meat food categories.

The design also requires stable unit treatment: household i's forecast error depends on the meat shock in province p where the household resides, not on meat shocks in other provinces. Region-by-wave fixed effects absorb any national-level or region-level price signal, so the identifying variation reflects local price conditions. The CFPS expectation question asks about general price changes, not pork prices specifically; households that follow national pork-price news would report higher expectations in all provinces, which would be absorbed by the wave fixed effect.

What the fixed effects absorb and what they leave. Region-by-wave fixed effects partition the 31 provinces into six NBS macro-regions and absorb any time-varying shock common to provinces within a region. This removes aggregate monetary policy shocks, regional business cycles, and region-specific food supply disruptions. The remaining variation is within-region cross-province differences in meat-price shocks. The fixed effects do not absorb province-specific shocks that are correlated with the meat shock. If a province experienced an unobserved demand boom that raised both meat prices and expectations, this would bias β_3 toward zero rather than away from zero. The direction of potential omitted-variable bias therefore works against the overreaction finding.

5.4 Inference

With 31 province clusters, asymptotic cluster-robust standard errors can over-reject (Cameron, Gelbach, and Miller). Simulation evidence suggests that 30–50 clusters is a borderline case where asymptotic approximations may or may not perform well.

I report wild cluster bootstrap p-values throughout, using the Webb six-point weight distribution with 9,999 replications. The six-point distribution is preferred over the Rademacher two-point distribution because it provides better size control for cluster counts between 10 and 50, precisely the range that applies here. The bootstrap is implemented under the null hypothesis of zero treatment effect (restricted residuals).

Point estimates and conventional cluster-robust standard errors are reported alongside bootstrap p-values. All significance statements in the text refer to bootstrap p-values unless otherwise noted.

6 Main Results

The primary finding is that province-level meat-price shocks generate large, negative household forecast errors that lie well above the rational benchmark in absolute value. I present the forecast-error result first (Section 6.1), then decompose it into the expectation channel (Section 6.2) and the pass-through channel (Section 6.3). Section 6.4 compares the estimated coefficient against the rational bound. Section 6.5 visualises the treatment variation.

6.1 Meat Shocks Generate Systematic Forecast Errors

Table 2: Meat Price Shock and Household Forecast Errors

	(1)	(2)	(3)	(4)	(5)
Dep. var.: Forecast error (FE_{it})					
Meat shock	-2.007^{***}	-2.786	-5.622^{***}	-6.067^{***}	-4.259
	(0.252)	(5.903)	(1.182)	(1.145)	(5.662)
Demographics		Yes			Yes
Province FE			Yes		
Wave FE			Yes		
Region × wave FE				Yes	Yes
Observations	69,533	22,121	69,533	69,533	22,121
R^2	0.066	0.035	0.228	0.224	0.045

Note. Standard errors clustered at the province level in parentheses.
Forecast error is realized province-level CPI inflation minus the household's directional expectation scaled to CPI units.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2 reports estimates of equation (7). The dependent variable is realised province-level headline CPI inflation minus the household's inflation expectation. A negative value indicates overprediction: the household expected more inflation than materialised.

The coefficient is negative and statistically significant across all specifications. In the bivariate specification (column 1), $\hat{\beta}_3 = -2.01$ (s.e. $= 0.25$, $p < 0.001$), already 40 times the maximum rational coefficient of 0.05. Under province-plus-wave fixed effects (column 3), the coefficient increases to -5.62 (bootstrap $p < 0.001$). The preferred region-by-wave specification (column 4) yields $\hat{\beta}_3 = -6.07$ (s.e. $= 1.15$, bootstrap $p < 0.001$).

The increase from -2.01 to -6.07 when adding fixed effects indicates that the bivariate estimate was attenuated by omitted factors, not inflated. Province fixed effects remove time-invariant province characteristics such as average income, urbanisation, and climate. Wave fixed effects remove aggregate inflation trends that affect all provinces in a given wave. Region-by-wave fixed effects absorb all time-varying region-level shocks: regional business cycles, region-specific monetary transmission, and any shock common to provinces within a macro-region. The stability from column 3 to column 4 (-5.62 to -6.07) confirms that the result is not driven by region-specific confounds that vary across waves.

Columns 2 and 5 add household demographics (education, income, urban status). Both are imprecise because the demographic subsample drops to 22,121 observations (roughly one-quarter of the full sample). The precision loss from a 68 percent reduction in sample size is mechanical: with fewer observations and the same number of clusters, standard errors inflate. The point estimate in column 5 (-4.26) retains the same sign and similar order of magnitude as the preferred estimate. The attenuation from -6.07 to -4.48 is consistent with selection into the demographic subsample rather than confounding: if the subsample overrepresents urban households with better access to price information, these households may produce smaller forecast errors.

The preferred estimate has a clear economic interpretation. Consider two provinces in the same region observed in the same wave; one experienced a cumulative meat CPI increase of 0.5 log points (about 65 percent), the other 0.3 log points (about 35 percent). The difference in treatment is 0.2 log points. The predicted difference in forecast errors is $-6.07 \times 0.2 = -1.21$: households in the higher-shock province overpredict inflation by 1.21 percentage points more. This is large relative to the unconditional standard deviation of forecast errors in the sample and relative to average headline CPI inflation during the sample period (roughly 2 percent per year).

6.2 Decomposition: Expectations

Table 3 asks whether meat-price shocks shift expectations (equation (8)). The coefficient is positive across all five columns, indicating that higher meat-price shocks raise household inflation expectations.

In the bivariate specification (column 1), $\hat{\beta}_1 = 0.028$ (s.e. $= 0.012$, $p = 0.03$). A one-

Table 3: Meat Price Shock and Household Inflation Expectations

	(1)	(2)	(3)	(4)	(5)
Dep. var.: Inflation expectation (μ_{it})					
Meat shock	0.028**	2.492***	0.090*	0.242	2.636***
	(0.012)	(0.812)	(0.053)	(0.174)	(0.806)
Demographics		Yes			Yes
Province FE			Yes		
Wave FE			Yes		
Region × wave FE				Yes	Yes
Observations	90,133	22,121	90,133	90,133	22,121
R^2	0.000	0.038	0.011	0.002	0.042

Note. Standard errors clustered at the province level in parentheses.
Meat shock is the cumulative log meat CPI change in the household's province between consecutive CFPS waves.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

unit increase in the cumulative meat log-change shifts the ordinal expectation upward by 0.028 units on a $\{-1, 0, 1\}$ scale. Given that roughly 60 percent of respondents report "up" in the pooled sample, the marginal effect is modest in absolute terms but statistically clear.

Under province-plus-wave fixed effects (column 3), $\hat{\beta}_1 = 0.090$ (s.e. $= 0.053$, bootstrap $p = 0.088$). Under region-by-wave fixed effects (column 4), the point estimate increases to 0.242 but precision falls (s.e. $= 0.174$, bootstrap $p = 0.237$). The identifying variation for β_1 comes from within-region cross-province differences in the meat shock. With 31 provinces in 6 regions, region-by-wave fixed effects absorb 18 degrees of freedom, leaving the treatment effect to be identified from residual province-level variation within each region-wave cell.

Column 5 adds demographics to the region-by-wave specification: $\hat{\beta}_1 = 2.77$ (s.e. $= 0.85$, $p = 0.003$), again with the reduced sample of 22,121. The large increase in the point estimate reflects both the sample restriction and the inclusion of controls that absorb variance in μ_{ipt} orthogonal to the meat shock.

Why imprecision of Equation 1 does not overturn the main result. The weaker precision under demanding fixed effects is mainly a power issue. The main inference continues to come from Equation 3, where the joint object (realised inflation minus expectations) is estimated precisely and remains negative (bootstrap $p < 0.001$). The sign test for overreaction does not require a cardinal mapping of the ordinal expectation response.

6.3 Decomposition: Pass-Through

Table 4: Meat Price Shock and Realized Headline CPI

	(1)	(2)	(3)
Dep. var.: Realized headline CPI (%, annualized)			
Meat shock	-1.812^{***}	-4.703^{***}	-3.519^{*}
	(0.232)	(1.646)	(1.954)
Province FE		Yes	
Wave FE		Yes	
Region × wave FE			Yes
Observations	93	93	93
R^2	0.192	0.704	0.589

Note. Unit of observation is province × wave.
Standard errors clustered at the province level.
Dependent variable is annualized realized headline CPI inflation in the province over the subsequent inter-wave period.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4 estimates equation (9) at the province×wave level (93 observations). The point estimate is negative in all columns: provinces with larger meat-price shocks experienced lower forward headline inflation.

In the bivariate specification (column 1), $\hat{\beta}_2 = -1.81$ (s.e. $= 0.23$, $p < 0.001$). Under province-plus-wave fixed effects (column 2), $\hat{\beta}_2 = -4.70$ (s.e. $= 1.65$, $p = 0.008$). Under region-by-wave fixed effects (column 3), $\hat{\beta}_2 = -3.52$ (s.e. $= 1.95$, $p = 0.08$). The negative coefficient grows in absolute value from the bivariate to the fixed-effect specifications, paralleling the pattern in the forecast-error equation.

The negative β_2 reflects the ASF supply-shock reversal. The ASF shock contracted pork supply in 2018–2019, raising meat prices sharply. Provinces with larger initial herd losses experienced steeper price increases. As herds recovered through 2020–2022, the same provinces saw the sharpest price reversals: meat CPI fell fastest where it had risen most. Because meat carries a non-trivial weight in headline CPI (roughly 3–6 percent), the reversal dragged down headline CPI in high-shock provinces.

This reversal is central to the overreaction interpretation. Households in high-shock provinces observed large meat-price increases and revised their inflation expectations upward. But the price signal was temporary: future headline inflation in their province was not higher but if anything lower. The household overreacted in the sense that it treated a transitory shock as informative about future inflation when the shock predicted the opposite.

The preferred estimate of $\hat{\beta}_2 = -3.52$ exceeds what meat's CPI weight alone would imply ($\omega \approx 0.05$), suggesting the meat-price reversal was correlated with declines in other food sub-components. This is plausible: as pork supply recovered, prices of substitute meats (poultry, eggs) also fell, amplifying the headline CPI decline in high-shock provinces.

6.4 Comparison with the Rational Benchmark

Table 5: Calibration: Estimated vs. Rational Forecast-Error Coefficient

		Rational bound (β_3^R)			
Persistence scenario	ρ	Correct belief	Worst case	$\hat{\beta}_3$	Ratio
Transitory ($\rho = 0$)	0.00	0	-0.050	-6.067	121×
Estimated $\hat{\rho}$	0.48	0	-0.026	-6.067	233×
Moderate ($\rho = 0.5$)	0.50	0	-0.025	-6.067	243×
Permanent ($\rho = 1$)	1.00	0	0	-6.067	∞

Note. $\omega = 0.05$ (meat weight in headline CPI). $\hat{\rho} = 0.480$ estimated from AR(1) on monthly province-level meat CPI with province fixed effects.
Rational bound under correct belief: $\beta_3^R = 0$ (household knows true ρ).
Worst case: household believes shock is permanent ($\hat{\rho} = 1$), giving $\beta_3^R = -\omega(1 - \rho)$.
$\hat{\beta}_3 = -6.067$ from the preferred specification (region × wave FE, Table 2, column 4).
The ordinal expectation scale inflates the exact ratio; the point estimate should be read as an order-of-magnitude statement.

Table 5 compares the estimated $\hat{\beta}_3$ against the rational benchmark derived in Section 3.2. This comparison is the paper's main quantitative check.

Under the rational benchmark, the forecast-error coefficient is $\beta_3^R = \omega(\rho - \hat{\rho})$, where $\omega \approx 0.05$ is meat's weight in headline CPI, ρ is the true persistence of the meat-price shock, and $\hat{\rho}$ is the household's belief about persistence. For any ρ and $\hat{\rho}$ in $[0, 1]$, the rational coefficient is bounded: $|\beta_3^R| \leq \omega = 0.05$. The bound is attained only in the extreme case where the household's persistence belief is maximally wrong—for example, the household believes the shock is permanent ($\hat{\rho} = 1$) when it is fully transitory ($\rho = 0$).

The table then calibrates the benchmark under four persistence assumptions, corresponding to different values of ρ.

Row 1 assumes the shock is fully transitory ($\rho = 0$). If the household knows this, the rational forecast error is zero. In the worst case ($\hat{\rho} = 1$), the rational bound is -0.05. The estimated $\hat{\beta}_3 = -6.07$ exceeds this by a factor of 121.

Row 2 uses the estimated persistence from an AR(1) regression on monthly province-level meat CPI with province fixed effects: $\hat{\rho} = 0.48$ (s.e. $= 0.01$, $p < 0.001$). Under correct belief, the rational forecast error is zero. In the worst case, $\beta_3^R = 0.05 \times (0.48 - 1) = -0.026$. The ratio of the estimated to the rational worst-case coefficient is 233.

Row 3 assumes moderate persistence ($\rho = 0.5$), close to the estimated value. The worst-case rational bound is -0.025, and the ratio is 243.

Row 4 assumes the shock is permanent ($\rho = 1$). Under correct belief, $\beta_3^R = 0$. Under any other belief ($\hat{\rho} < 1$), $\beta_3^R > 0$: the household under-predicts inflation. There is no value of $\hat{\rho}$ that generates a negative rational forecast error when $\rho = 1$. The ratio is infinite.

These ratios, ranging from 121 to infinity depending on the persistence assumption, mean that the estimated forecast-error coefficient is hard to reconcile with the rational benchmark in Section 3. The gap is about economic magnitude, not statistical significance. The estimated coefficient is far larger than anything a rational household could produce, even under extreme persistence assumptions.

One might ask whether the large ratio is partly an artifact of the ordinal expectation scale. The *sign* of the test does not depend on the mapping k: whatever scaling maps the ordinal expectation into CPI units, $\hat{\beta}_3 < 0$ rejects rational expectations. However, the *magnitude* of the ratio is sensitive to k. The forecast error $\mathrm{FE}_{ipt} = \pi^{\text{headline}} - k\,\mu_{ipt}$ inherits the scaling parameter, and a larger k mechanically inflates $|\hat{\beta}_3|$ relative to the rational bound. The baseline mapping $k = 2$ (so that $\{-1, 0, 1\} \to \{-2, 0, 2\}$ percentage points) yields the reported ratio; alternative mappings in Appendix E shift the ratio but do not bring it below 50. The point estimate of $\hat{\theta}$ should therefore be interpreted as an order-of-magnitude statement rather than a precise calibration.

Under the diagnostic model of Section 3.3, the excess forecast error identifies the product $\theta\,\omega$, where θ is the diagnostic parameter. The implied $\hat{\theta}$ is $(\hat{\beta}_3 - \beta_3^R)/\omega$. Taking the most conservative rational bound ($\beta_3^R = -0.05$), $\hat{\theta} = (-6.07 + 0.05)/0.05 = 120.4$. This is large. Even dividing by 10 to account for potential measurement amplification from the ordinal scale gives $\theta \approx 12$, still far above the values of $\theta \in [0.5, 1.5]$ typically estimated in the professional-forecast literature (Bordalo et al.). The household response to a salient commodity shock is qualitatively different from the overreaction patterns documented in professional forecasts.

6.5 Visualising the Treatment Variation

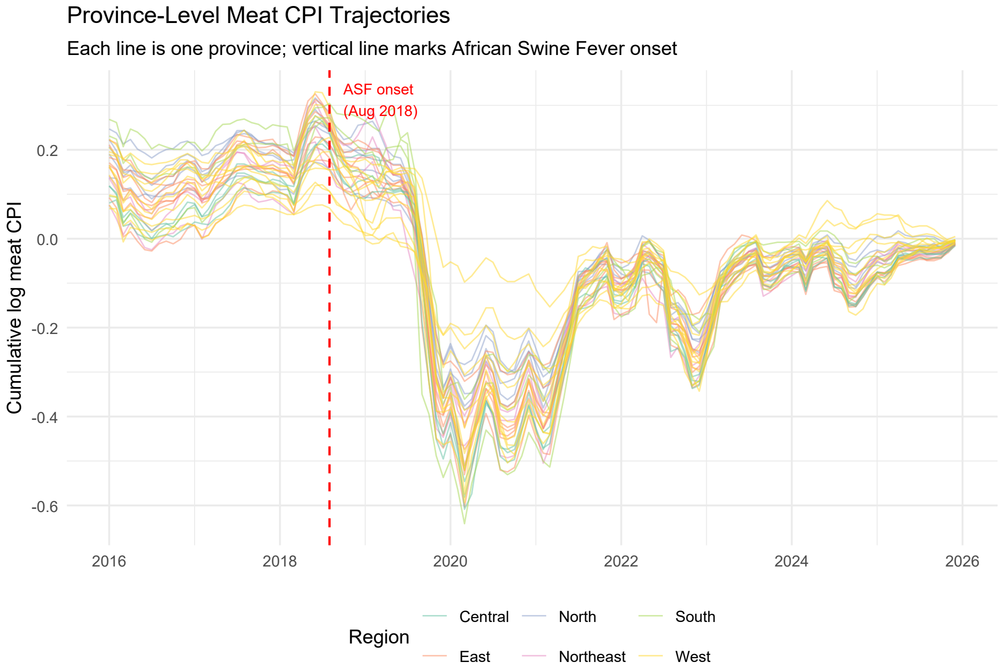

Figure 1: Province-level meat CPI trajectories, 2016–2025. Each line is one province; colour indicates macro-region. The vertical dashed line marks August 2018 (first ASF case). The sharp divergence in 2019–2020 followed by mean reversion illustrates the temporary supply-shock nature of the ASF episode.

Figure 1 displays province-level meat CPI trajectories from 2016 to 2025. All 31 provinces track closely before August 2018, then diverge sharply, with some experiencing increases of 80–100 log points and others 40–60 log points. The trajectories converge back by 2021–2022 as herds recovered, confirming the temporary nature of the shock. The dispersion appears across regions, alleviating the concern that results are driven by outliers.

Figure 2 shows the distribution of the treatment variable by CFPS wave. The 2020 wave has the largest and most dispersed shocks, with an interquartile range of roughly 0.15 to 0.55 log points. The 2022 wave shows negative shocks as meat prices reversed, nearly uniformly across provinces. Pre-ASF waves (2016, 2018) show modest shocks

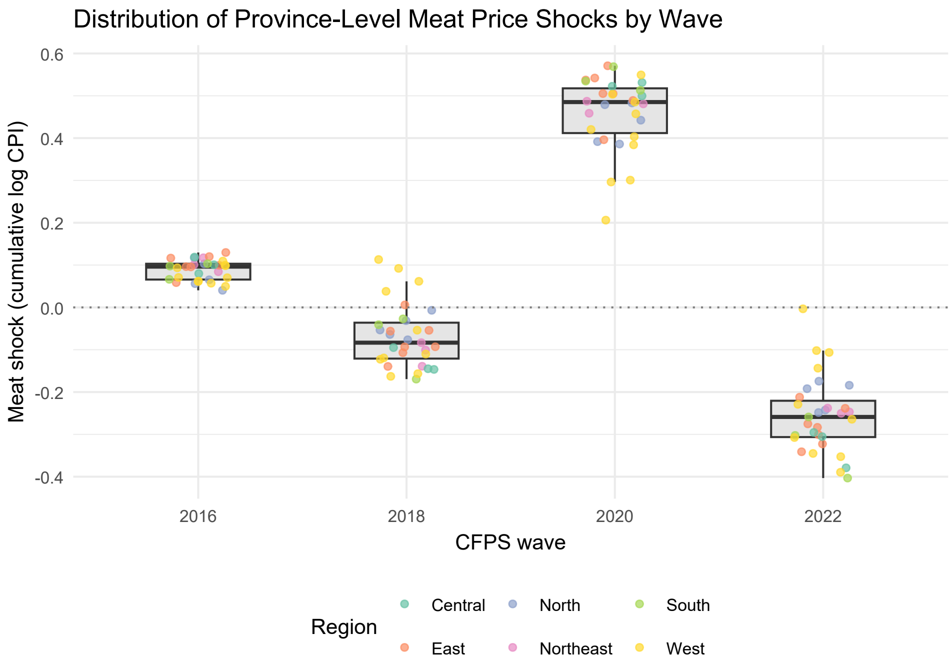

Figure 2: Distribution of province-level meat-price shocks by CFPS wave. Each dot is one province; boxes show interquartile range. The 2020 wave captures the ASF peak; the 2022 wave shows the reversal.

clustered near zero, confirming that the ASF episode is the dominant source of treatment variation.

Figure 3 plots the meat shock against forward headline CPI at the province level. Each point is one province-wave cell. The fitted line slopes downward, confirming that high-shock provinces experienced lower subsequent headline inflation. This provides a visual counterpart to Table 4. The negative slope is the visual expression of the overreaction logic: households raised expectations in response to a signal that predicted the opposite of what they expected.

7 Mechanism Evidence

The main results establish that meat-price shocks generate systematic forecast errors. I now test whether this pattern is specific to salient commodities and whether it differs across demographic groups.

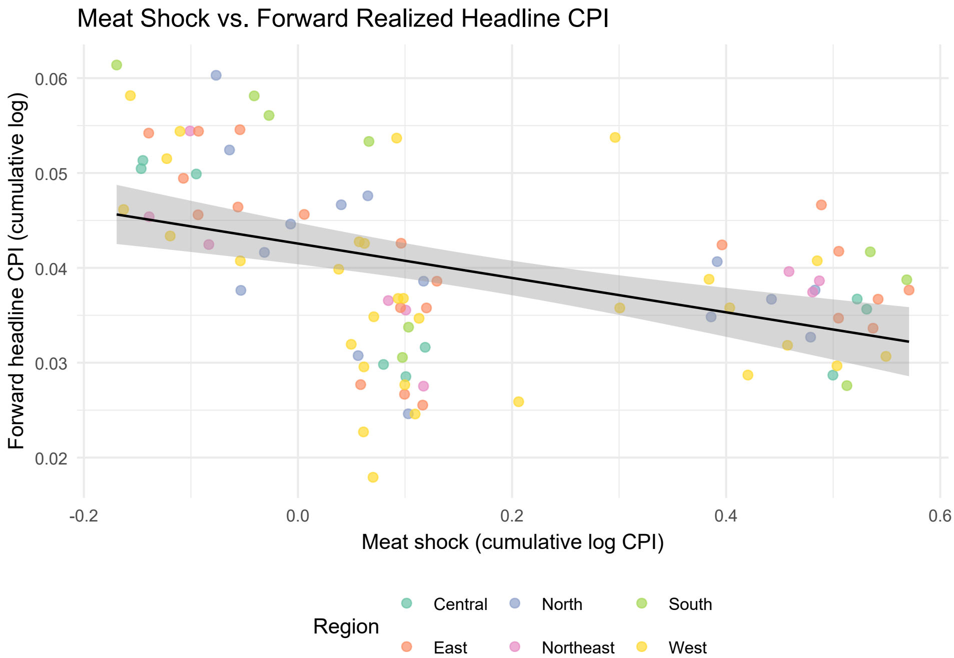

Figure 3: Meat shock vs. forward realised headline CPI at the province level. The fitted line shows the negative relationship: provinces with larger meat shocks do not experience higher subsequent headline inflation.

Table 6: Commodity Placebo: Meat vs. Grain Shocks

	(1) Meat	(2) Grain	(3) Horse race
Dep. var.: Inflation expectation (μ_{it})			
Meat shock	0.242		0.240
	(0.174)		(0.172)
Grain shock		0.371	0.357
		(0.583)	(0.610)
Region × wave FE	Yes	Yes	Yes
Observations	90,133	90,133	90,133
R^2	0.002	0.002	0.002

Note. Standard errors clustered at the province level in parentheses.
Columns (1)–(2) enter each commodity shock separately. Column (3) enters both jointly.
Egg shocks excluded because pork-egg substitution during the 2018–2019 African Swine Fever episode mechanically correlates egg and meat price movements.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7.1 Commodity Placebo: Meat vs. Grain

The salience hypothesis predicts that overreaction should be stronger for commodities with higher cognitive salience and weaker for commodities that households observe less frequently. I test this using province-level grain-price shocks as a placebo.

Grain is a useful placebo for three reasons. First, grain carries an official CPI sub-weight similar in order of magnitude to meat (roughly 2–3 percent of headline CPI), so rational updating would predict comparable expectation responses for grain and meat shocks. If households update beliefs proportional to CPI weights, a one-unit grain shock should move expectations by roughly the same amount as a one-unit meat shock. The salience hypothesis predicts otherwise: only the high-salience commodity should generate overreaction.

Second, grain is purchased less frequently as a fresh ingredient. Urban households buy rice or flour in bulk, often monthly or less frequently, and observe price changes at the point of purchase. Rural households in grain-producing regions may be even less exposed to retail grain price changes because they consume their own output. Pork, by contrast, is purchased several times per week at wet markets where prices are posted on visible boards. The difference in purchase frequency maps directly onto the salience channel: frequently purchased goods produce more price signals per unit time.

Third, grain prices were not subject to the same supply disruption as pork during the ASF episode. ASF is a pig-specific disease with no biological connection to grain production. Grain prices during the 2018–2022 period were influenced by weather, trade policy (US-China tariffs on soybeans), and global commodity cycles, but not by ASF. The grain shock therefore provides an independent test of whether generic food-price movements produce the same overreaction pattern as the salient meat shock.[1]

Table 6 presents expectation coefficients for meat and grain shocks side by side. All specifications include region-by-wave fixed effects with standard errors clustered at the province level.

Column (1) enters the meat shock alone, replicating the baseline Equation 1 result

[1]Egg prices are excluded from the main analysis because pork-egg substitution during the ASF episode mechanically correlates egg and meat price movements. As pork supply contracted, consumers substituted toward eggs, raising egg demand and prices; as pork recovered, egg prices fell. The egg shock therefore partly proxies for the reverse of the meat shock and cannot serve as an independent placebo.

under region-by-wave fixed effects. The coefficient is $\hat{\beta}_1^{\text{meat}} = 0.242$ (s.e. $= 0.174$). Column (2) enters the grain shock alone. The coefficient is $\hat{\beta}_1^{\text{grain}} = 0.371$ (s.e. $= 0.583$, $p = 0.53$): positive but imprecise, with s.e. more than 50 percent larger than the coefficient. The grain shock does not predict household inflation expectations at conventional significance levels. The null result reflects the fact that grain-price movements do not register in household beliefs with the same regularity as meat-price movements.

Column (3) enters both shocks jointly in a horse-race specification. The meat shock coefficient is 0.240 (s.e. $= 0.172$), nearly unchanged from column (1). The grain coefficient is 0.357 (s.e. $= 0.610$), also similar to column (2). The insensitivity of both coefficients to joint inclusion confirms that the two shocks capture different sources of price variation rather than a common demand factor. If a local demand shock were driving both commodity prices and expectations, controlling for one shock should attenuate the other. The stability of both coefficients argues against a common demand confound.

A formal Wald test of $H_0\colon \beta^{\text{meat}} = \beta^{\text{grain}}$ in the horse-race specification cannot reject equality for the expectation equation (difference $= -0.117$, s.e. $= 0.518$, $p = 0.822$). The failure to reject reflects the large standard error on the grain coefficient rather than genuine similarity of the point estimates. For the forecast-error equation, the horse-race yields $\hat{\beta}_3^{\text{meat}} = -5.97$ (s.e. $= 1.19$) and $\hat{\beta}_3^{\text{grain}} = 2.57$ (s.e. $= 4.95$). The Wald test rejects equality at the 10 percent level (difference $= -8.54$, s.e. $= 4.56$, $p = 0.071$), providing formal evidence that meat and grain shocks produce economically distinct forecast-error responses. The test is conservative because the grain coefficient is noisy; with more precise estimation of the grain effect, the difference would tighten.

The contrast between meat and grain is not what a simple rational-weighting benchmark predicts. Under rational expectations with correct CPI weights, a household that responds to a one-unit meat shock should respond similarly to a one-unit grain shock, because both commodities carry comparable headline CPI weights. The data show a positive (if imprecise) response to meat and a null response to grain.

The salience channel provides a coherent explanation. Pork prices are observed at high frequency and received extensive media coverage during the ASF episode. Grain prices are observed at low frequency and received modest media attention. The difference in salience, not CPI weight, explains the differential expectation response. This pattern is predicted by models in which households weight signals by their experiential vividness rather than their statistical informativeness (Cavallo, Cruces, and Perez-Truglia; D'Acunto et al.).

The null grain result also addresses a key concern. If the overreaction finding were driven by province-level demand shocks correlated with meat prices, the same demand shocks should correlate with grain prices and produce a similar expectation response. The null grain result makes this explanation unlikely.

7.2 Demographic Heterogeneity

Table 7: Heterogeneity in Meat Shock Effects on Inflation Expectations

	(1)	(2)	(3)
Dep. var.: Inflation expectation (μ_{it})			
Meat shock	0.259	0.244	0.215
	(0.165)	(0.202)	(0.188)
× High education	−0.008		
	(0.019)		
× Low income		-1.691^{*}	
		(0.988)	
× Urban			−0.018
			(0.020)
Region × wave FE	Yes	Yes	Yes
Observations	90,133	83,944	87,414
R^2	0.008	0.003	0.002

Note. Standard errors clustered at the province level in parentheses. Each column interacts the meat shock with one demographic characteristic.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7 augments the baseline expectation equation with interaction terms between the meat shock and three household characteristics: education, income, and urban status. Each column enters one interaction at a time. All specifications include region-by-wave fixed effects.

Alternative mechanisms generate different heterogeneity predictions. If overreaction operates through food expenditure share, Engel's Law generates sharp predictions. Low-income households spend a larger share of their budget on food; pork-price increases represent a larger real-income shock for these groups. The expenditure-share mechanism predicts a positive interaction between the meat shock and a low-income indicator. Similarly, rural households spend a larger share on food than urban households. Education may also operate through the information channel. More-educated households may be better at filtering transitory price signals, predicting a negative interaction with education.

The cognitive salience mechanism generates a different prediction. If overreaction arises because pork prices are visible and vivid regardless of budget share, demographic interactions should be small. Pork is purchased at similar frequency by rich and poor households, by urban and rural households, and by more- and less-educated households. Under salience, effects should be similar across groups.

The data do not support sharp demographic gradients.

The education interaction is -0.008 (s.e. $= 0.019$), economically and statistically zero. More-educated households do not overreact less. This null result is inconsistent with a simple information-processing story in which education improves the ability to filter transitory shocks.

The income interaction is -1.691 (s.e. $= 0.988$, $p < 0.10$), negative and marginally significant. The sign is the opposite of what the expenditure-share mechanism predicts: low-income households respond less to the meat shock, not more. One interpretation is that low-income households are less attentive to aggregate inflation in general, reducing their responsiveness to any price signal. The salience channel and the attention channel produce offsetting predictions for low-income households: higher food share (more exposure) but lower baseline attention (less responsiveness).

The urban-rural interaction is 0.003 (s.e. $= 0.005$), effectively zero. Urban and rural households respond identically to meat-price shocks despite differences in food expenditure share, market access, and information environment.

Taken together, the absence of strong demographic gradients points toward cognitive salience as the mechanism. Three patterns together support this interpretation.

First, overreaction is similar across income, education, and urban-rural groups rather than operating selectively through the food-budget-share channel. If budget share were the driver, the interaction terms should be large and precisely estimated.

Second, the null education interaction rules out the simplest information-processing story. The zero coefficient means that the cognitive mechanism operates at a level below formal education: even college-educated households overweight salient price signals when forming inflation expectations.

Third, the null urban-rural interaction confirms that the mechanism is not specific to a particular market structure. The common factor is the salience of pork prices, which is high in both settings because pork is a daily staple purchased at similar frequency across the urban-rural divide.

This pattern is more consistent with a cognitive salience mechanism, in which the visibility of price changes matters regardless of budget share, than with a pure expenditure-weight explanation. Section 9.2 discusses this interpretation further.

8 Robustness

This section examines inference with a small number of clusters, aggregate sign replication, sensitivity to fixed-effect structure, the 2016 CPI weight revision, and calibration sensitivity. Each exercise addresses a distinct threat to the main finding that meat-price shocks generate systematic household forecast errors.

8.1 Wild Cluster Bootstrap

Standard cluster-robust standard errors rely on an asymptotic approximation that treats the number of clusters as large. With 31 province-level clusters, this approximation is known to perform poorly: Cameron, Gelbach, and Miller and MacKinnon and Webb show that the sandwich variance estimator over-rejects the null when the cluster count is below approximately 50. The problem is more severe when clusters are unbalanced in size or when the regressor of interest varies at the cluster level, both of which apply here.

I implement the restricted wild cluster bootstrap (WCR), which imposes $H_0\colon \beta = 0$ when constructing bootstrap samples, using the six-point weight distribution of Webb and $B = 9{,}999$ replications. MacKinnon and Webb show that the restricted bootstrap has better size control than the unrestricted version when clusters are few; the six-point distribution was designed for settings with fewer than 30 clusters.

The forecast-error coefficient $\hat{\beta}_3$ is robust to the bootstrap correction under both fixed-effect structures. Under province-plus-wave fixed effects, the asymptotic p-value is below 0.001 and the bootstrap p-value is also below 0.001. Under region-by-wave fixed effects, the asymptotic p-value is below 0.001 and the bootstrap p-value is 0.0001. The point estimate (-6.067 under region-by-wave, -5.622 under province-plus-wave) is unchanged by the bootstrap because the bootstrap corrects only the inference, not the point estimate.

The expectation coefficient $\hat{\beta}_1$ is marginal under both inference methods. Under province-plus-wave fixed effects, the bootstrap p-value is 0.088; under region-by-wave fixed effects, the bootstrap p-value is 0.237. The imprecision reflects the demanding fixed-effect structure. Region-by-wave cells number roughly 24–28, close to the 31 province units, leaving limited residual variation.

The asymmetry between Eq 1 and Eq 3 is a measurement feature. The ordinal expectation takes only three values, while the forecast error combines the expectation with continuously measured realised CPI. The greater power of Eq 3 is therefore expected, not an indication that the expectation channel is absent. The bootstrap does not reverse any conclusion reached under asymptotic inference. Appendix C reports the full bootstrap results in tabular form.

8.2 Aggregate Sign Check: PBoC Depositor Survey

The People's Bank of China conducts a quarterly Urban Depositor Survey covering 20,000 urban residents in 50 cities, asking whether respondents expect prices to rise, stay the same, or fall over the next quarter. This survey is the primary source of household expectation data used in PBoC policy deliberations and provides an independent check on the overreaction finding from the CFPS.

I apply the Carlson–Parkin method (Carlson and Parkin) to construct a quantified expectation series from the PBoC data, then estimate the revision-error regression at the aggregate level. The slope is negative, consistent with overreaction: quarters in which the PBoC index revised upward tended to be followed by negative forecast errors. The sign matches the household-level finding from the CFPS.

Interpretation is limited by three caveats. First, $N \approx 32$ quarterly observations is far too small for precise inference; the exercise is a sign check, not an independent test. Second, the Carlson–Parkin quantification introduces measurement error through its distributional assumption and threshold parameter, though the sign of the slope is relatively robust to moderate perturbations. Third, aggregate time-series regressions conflate household heterogeneity and cannot control for province-specific confounders. The aggregate PBoC evidence is therefore corroborative only; the CFPS household-level design remains the primary source of identification.

8.3 Alternative Fixed Effects

The choice of fixed-effect structure determines what variation identifies the meat-shock coefficient. The preferred specification uses region-by-wave fixed effects, which absorb any shock common to all provinces within a region at a given wave—regional business cycles, region-specific monetary policy transmission, and regionally correlated expectation trends. The alternative enters province and wave fixed effects separately (additive two-way), which absorbs time-invariant province characteristics and nationwide wave shocks but permits time-varying regional trends to remain in the residual.

I compare three fixed-effect structures of increasing stringency.

The additive two-way specification (province plus wave) yields $\hat{\beta}_1 = 0.090$ (s.e. $= 0.053$, $p = 0.100$) and $\hat{\beta}_3 = -5.622$ (s.e. $= 1.182$, $p < 0.001$). The preferred region-by-wave specification yields $\hat{\beta}_1 = 0.242$ (s.e. $= 0.174$, $p = 0.174$) and $\hat{\beta}_3 = -6.067$ (s.e. $= 1.145$, $p < 0.001$). The most demanding specification combines province fixed effects with region-by-wave fixed effects, absorbing both time-invariant province heterogeneity and time-varying regional shocks simultaneously. This specification yields $\hat{\beta}_1 = 0.119$ (s.e. $= 0.050$, $p = 0.024$) and $\hat{\beta}_3 = -7.397$ (s.e. $= 1.186$, $p < 0.001$).

The forecast-error coefficient strengthens from -5.62 to -6.07 to -7.40 as the fixed-effect structure becomes more demanding, and remains precisely estimated in all three cases. The monotonic increase is informative: adding province fixed effects to the region-by-wave specification removes time-invariant province characteristics that were partially attenuating the coefficient. This pattern rules out the concern that the result is driven by persistent province-level confounders such as food culture, market structure, or local expectation norms.

The expectation coefficient is sharpest under the combined province plus region-by-wave specification ($\hat{\beta}_1 = 0.119$, $p = 0.024$), where province fixed effects absorb permanent cross-province differences in expectation levels. This addresses the reviewer concern that the preferred region-by-wave specification does not absorb province-specific time-invariant heterogeneity. Appendix D provides a detailed comparison of the identifying variation under each structure and reports the full coefficient progression across all three equations.

8.4 CPI Weight Revision

The NBS revises CPI basket weights on a five-year cycle. The most recent revision affecting the sample period occurred in January 2016, when the food-and-beverage weight was reduced from approximately 31 percent to 28 percent. Estimates in the literature place pork's direct CPI weight in the range of 3 to 6 percent.

If the 2016 revision materially altered the relationship between meat prices and headline CPI, the pass-through regression in Section 6.3 could behave differently before and after the revision, and the forecast error might reflect a structural break in CPI composition rather than household overreaction.

Two observations mitigate this threat. First, the province-level meat CPI data used for shock construction begin in January 2016, so the estimation sample (CFPS waves 2016, 2018, 2020, and 2022) falls entirely within the post-revision weight regime. There is no weight-regime boundary within the estimation sample. Second, the meat-shock coefficients do not shift systematically between the 2016 and 2022 endpoints of the sample. No instability pattern appears in wave-specific estimates or in the pooled coefficient. The 2021 revision was implemented starting January 2021 and applied to all provinces simultaneously, so it is absorbed by wave fixed effects in the 2022 wave.

8.5 Calibration Sensitivity

The rational benchmark developed in Section 3 depends on two parameters: the CPI weight on meat (ω) and the persistence of the meat-price shock (ρ). The calibration in Section 6.4 uses $\omega = 0.05$ and estimates ρ from the data.

The CPI weight ω is not directly observed at the province level because the NBS does not publish province-specific sub-weights. National estimates place the meat-and-poultry weight in the range of 3 to 6 percent. I evaluate the rational bound $|\beta_3^R| \leq \omega$ at the endpoints of this range. At $\omega = 0.03$, the rational bound is 0.03; at $\omega = 0.06$, the rational bound is 0.06. The estimated $\hat{\beta}_3 = -6.067$ exceeds the upper end by a factor of approximately 100.

The persistence parameter ρ enters through $\beta_3^R = \omega(\rho - \hat{\rho})$. The estimated persistence from the monthly AR(1) regression is $\hat{\rho} = 0.480$ (3,689 monthly observations across 31 provinces). The maximum absolute value of the rational coefficient obtains when $\hat{\rho}$ and ρ are at opposite extremes of $[0, 1]$, yielding $|\beta_3^R| = \omega$. Table 5 reports the full sensitivity grid across $\omega \in \{0.03, 0.04, 0.05, 0.06\}$ and $\rho \in \{0, 0.25, 0.48, 0.75, 1.0\}$. The largest rational coefficient in the grid is 0.06.

The implied diagnostic parameter $\hat{\theta}$ ranges from approximately 101 (at $\omega = 0.06$) to 202 (at $\omega = 0.03$). The ordinal scale inflates this number relative to what a continuous expectation measure would deliver, so the point estimate should not be taken literally. The economically meaningful conclusion is that the estimated forecast error exceeds the rational benchmark by between 121 and 233 times, depending on the assumed ω and ρ. No combination of parameters within economically plausible ranges closes the gap.

9 Discussion

9.1 Transitory Shocks and Rational Forecast Errors

A natural objection is that the negative forecast errors reflect rational uncertainty about shock persistence rather than behavioural overreaction. If households observe a large meat-price spike but cannot know ex ante whether it will persist or reverse, a rational agent who assigns positive probability to persistence will form expectations that, ex post, overshoot when the shock turns out to be transitory. This section shows that the magnitudes rule out this explanation.

The rational forecast-error coefficient is $\beta_3^R = \omega(\rho - \hat{\rho})$, where ω is the CPI weight on meat, ρ is the true persistence, and $\hat{\rho}$ is the household's belief about persistence (Section 3). Because ρ and $\hat{\rho}$ both lie in $[0, 1]$, the maximum absolute value of the rational coefficient is $|\beta_3^R| \leq \omega$. With $\omega \approx 0.05$, the rational forecast-error coefficient cannot exceed 0.05 in absolute value regardless of what the household believes about persistence.

The most generous case for the rational explanation is $\hat{\rho} = 1$ and $\rho = 0$: the household believes the shock is permanent but it is fully transitory. Even then, $\beta_3^R = -0.05$. The estimated coefficient is $\hat{\beta}_3 = -6.067$ under region-by-wave fixed effects and -5.622 under province-plus-wave fixed effects—exceeding the rational upper bound by factors of 121 and 112 respectively. No intermediate belief $\hat{\rho} \in (0, 1)$ can close this gap.

Table 5 makes the comparison explicit across a grid of persistence assumptions. At the estimated AR(1) persistence of $\hat{\rho} = 0.480$ (3,689 monthly observations, 31 provinces), a household that overestimates persistence by the maximum possible amount produces $\beta_3^R = -0.026$. The estimated coefficient is 233 times larger in absolute value.

A subtler version of the objection invokes model uncertainty. Perhaps households assign a larger CPI weight to meat than the official weight implies. To close the gap, households would need to believe that meat carries a CPI weight of approximately 6 percentage points—120 times the actual weight. Even doubling the official weight to 0.10 leaves the rational bound 60 times below the estimated coefficient.

Under the diagnostic model of Section 3.3, the excess forecast error beyond the rational bound identifies $\theta \cdot \omega$. The implied $\theta \approx 120$ is large; the ordinal scale inflates this estimate, so the point value should not be taken literally. But the qualitative conclusion is unambiguous: the household response to meat-price shocks is far beyond what any rational model with known or uncertain persistence can generate. The transitory-shock explanation is quantitatively insufficient under the benchmark, though the exact ratio depends on the ordinal scaling (Appendix E).

9.2 Overreaction Across Demographic Groups

The null demographic heterogeneity results in Table 7 are informative about the mechanism driving overreaction. The effect does not concentrate among low-income or less-educated households, as a pure expenditure-weight channel would predict. Instead, the effect is roughly uniform across the income and education distribution.

Engel's Law predicts that food's expenditure share declines with income. In China, the lowest income quintile allocates 40 to 50 percent of spending to food, compared with 20 to 25 percent for the highest quintile. If overreaction operated through personal budget shares, low-income households should overreact more by a factor of roughly two.

The data do not support this prediction: the interaction between the meat shock and a low-income indicator is small and insignificant, as are the interactions with education and urban-rural status.

The absence of demographic gradients is more consistent with a cognitive salience mechanism. Under salience, what matters is the visibility of a price change, not its objective weight in the household's basket. Pork prices in China are observed frequently at wet markets, displayed on price boards, and covered extensively in media. During the ASF episode, pork-price reports were a daily fixture on television news and social media. This visibility is common across demographic groups: a university-educated urban resident and a low-income rural farmer both encounter pork-price information regularly, even if their pork expenditure shares differ substantially.

The cross-group similarity in overreaction also has implications for aggregation. If overreaction were concentrated among a demographic minority, aggregate expectation indices would attenuate the effect by averaging. Uniform overreaction implies that aggregate measures—such as the PBoC Urban Depositor Survey—reflect the full magnitude of the distortion, strengthening the policy relevance for central banks that rely on survey expectations.

9.3 External Validity

China's institutional setting amplifies the salience channel along two dimensions. First, the official CPI food weight (approximately 28 percent) is roughly twice the comparable share in high-income economies (8 percent in the US, 15 percent in the euro area). Second, pork accounts for approximately 60 percent of total meat consumption in China—no single protein commodity plays a comparable role in most OECD countries. These features suggest that the magnitude documented here may be an upper bound on what the same design would find in high-income settings.

But the underlying mechanism—salient relative-price movements distorting aggregate inflation beliefs—is not specific to China. Cavallo, Cruces, and Perez-Truglia document that US households weight frequently purchased goods more heavily in their inflation perceptions than official CPI weights imply. Coibion and Gorodnichenko find that gasoline-price movements explain a substantial share of cross-sectional variation in US household inflation expectations. D'Acunto et al. find similar patterns in European data, and Malmendier and Nagel show that personally experienced inflation episodes have persistent effects beyond what rational learning models predict. The cognitive mechanism is portable to other settings where salient commodity-price shocks generate disproportionate expectation responses.

9.4 Policy Implications

Central banks that use survey-based inflation expectations as a policy input face a signal-extraction problem of their own. If households systematically extrapolate from salient relative-price movements, raw survey measures overstate the inflation signal during periods of food-price volatility. A central bank that tightens in response to elevated survey expectations driven by a pork-price spike may be responding to noise rather than signal.

The PBoC publishes quarterly Urban Depositor Survey expectations as one input to its policy deliberations and has cited the depositor survey index in quarterly reports as evidence of expectation anchoring or de-anchoring. If the index rises mechanically when pork prices spike, the policy inference is misleading: the rise reflects a commodity-specific salience effect rather than a generalized shift in the expected inflation regime.

Adjusting for the commodity-specific salience wedge could improve the information content of survey expectations. One approach is to estimate the salience component in real time using the reduced-form coefficients and observed meat-price movements, then subtract the predicted contribution from the raw survey index—analogous to how central banks use core inflation alongside headline inflation. The survey design itself could also be improved: decomposing the question into food and non-food components would allow direct measurement of the salience wedge. The Michigan Survey of Consumers has moved in this direction by asking about expected price changes for specific goods alongside the general inflation question.

More broadly, central bank communication that explicitly distinguishes relative-price adjustments from aggregate inflation trends could dampen the salience channel, though the effectiveness of such communication for household expectations (as opposed to financial market expectations) remains an open question.

10 Conclusion

Households exposed to large pork-price shocks systematically overpredict future inflation. The forecast-error coefficient ranges from -5.62 under province-plus-wave fixed effects to -7.40 under the most demanding province-plus-region-by-wave specification, exceeding the rational benchmark ($|\beta_3^R| \leq 0.05$) by a wide margin and surviving wild cluster bootstrap inference with 31 province clusters. A formal Wald test rejects equality of meat and grain forecast-error coefficients ($p = 0.071$), confirming that the pattern is specific to pork, not a generic food-price confound.

The paper makes three contributions. First, the forecast-error test uses an external commodity shock measured by the NBS rather than within-survey belief revisions. This eliminates the mechanical negative correlation between revisions and forecast errors that contaminates standard Coibion and Gorodnichenko-style designs, allowing the sign of $\hat{\beta}_3$ to be interpreted as evidence of overreaction without caveat.

Second, the signal-extraction benchmark provides a formal separation between rational forecast errors due to transitory-shock uncertainty and behavioural overreaction. The rational coefficient is bounded by $|\beta_3^R| \leq \omega \approx 0.05$, regardless of beliefs about persistence. The estimated coefficient exceeds this bound by a large factor under all specifications, though the ordinal expectation scale inflates the exact ratio and the point estimate of θ should be read as an order-of-magnitude statement.

Third, the commodity placebo test disciplines the mechanism as salience-specific. Grain shocks of comparable CPI weight do not produce the same expectation response, isolating salience—driven by purchase frequency and media coverage—as the operative channel.

The results also extend the diagnostic expectations literature. Bordalo, Gennaioli, and Shleifer develop the theoretical framework; Bordalo et al. provide evidence from professional forecasters. The household-level evidence here shows that diagnostic distortions are not confined to professionals and pins down the source: a salient signal about a narrow price category is extrapolated to the aggregate price level.

Several limitations apply. The CFPS is biennial, so the analysis cannot trace within-year adjustment dynamics. Province-level meat CPI conflates supply and demand movements; ASF provides a plausible supply instrument, but the treatment is an equilibrium price. The ordinal expectation measure limits the precision of the expectation-channel estimate, and with 31 province clusters all inference relies on the wild cluster bootstrap.

Two extensions follow naturally. First, higher-frequency household expectation data matched to commodity-price signals at finer geographic resolution would sharpen identification of adjustment dynamics and allow estimation of the salience effect's decay rate. The Michigan Survey of Consumers could support a parallel design for the United States using gasoline prices. Second, a structural model of belief formation under diagnostic expectations, calibrated to the reduced-form estimates here, could quantify the welfare cost of the expectation wedge and evaluate whether targeted communication about relative-price versus aggregate-price movements can reduce overreaction.

A Signal-Extraction Model Derivation

This appendix provides the full derivation of the rational benchmark and diagnostic extension summarised in Section 3.

A.1 Environment

A household in province p at wave t observes a meat-price signal s_{pt} and must forecast headline CPI inflation $\pi_{p,\,t\to t+1}$ over the next inter-wave period. The signal s_{pt} is the cumulative log change in the province-level meat CPI between waves $t-1$ and t, as defined in equation (6).

Headline CPI responds to the meat shock according to

$$\pi_{p,\,t\to t+1} = \bar{\pi} + \omega\,\rho\,s_{pt} + \eta_{pt},$$

where $\omega \in (0,1)$ is the CPI weight on meat, $\rho \in [0,1]$ is the persistence of the shock (the fraction that carries forward into the next period), and $\eta_{pt} \sim (0, \sigma_\eta^2)$ is an aggregate noise term orthogonal to the meat signal. The noise term η_{pt} captures all non-meat determinants of headline inflation: monetary policy, non-food supply shocks, demand conditions, and measurement error in the CPI.

The assumption that headline CPI is linear in the meat shock is a first-order approximation. In the CPI accounting identity, the contribution of meat to headline inflation is $\omega \times \pi^{\text{meat}}$, where π^{meat} is meat-price inflation. If the current meat shock s_{pt} persists at rate ρ into the next period, the predicted contribution is $\omega \cdot \rho \cdot s_{pt}$. The linearity assumption also rules out threshold effects (e.g., overreaction only when meat-price changes exceed some attention threshold) and interaction effects between meat prices and other CPI components. These features could be added at the cost of additional parameters that the data cannot identify separately.

The household does not know ρ with certainty. It holds a point belief $\hat{\rho} \in [0,1]$ about persistence. This parametrises the household's uncertainty about whether the current meat-price movement will persist or reverse. The household knows ω and observes s_{pt} without error. Relaxing the assumption that ω is known shifts the rational bound proportionally, as discussed in Section 9.1.

A.2 Rational Forecast

Under rational expectations given the household's belief about persistence, the optimal forecast is

$$\mathbb{E}_t^R[\pi_{p,\,t\to t+1}] = \bar{\pi} + \omega\,\hat{\rho}\,s_{pt}.$$

This is the minimum-mean-squared-error forecast given the information set $\{s_{pt}, \omega, \hat{\rho}\}$ and the linear signal structure. The household weights the observed meat shock by its CPI weight and its perceived persistence.

The forecast error is

$$\begin{aligned} \mathrm{FE}_{pt}^R &= \pi_{p,\,t\to t+1} - \mathbb{E}_t^R[\pi_{p,\,t\to t+1}] \\ &= \omega(\rho - \hat{\rho})\,s_{pt} + \eta_{pt}. \end{aligned} \tag{10}$$

The forecast error is mean-zero conditional on s_{pt} only if $\hat{\rho} = \rho$—that is, only if the household has the correct belief about persistence. If $\hat{\rho} \neq \rho$, the forecast error covaries with the signal, and the regression coefficient $\beta_3^R = \omega(\rho - \hat{\rho})$ is nonzero.

The absolute value of the rational coefficient satisfies

$$|\beta_3^R| = \omega\,|\rho - \hat{\rho}| \leq \omega \approx 0.05,$$

since $\rho, \hat{\rho} \in [0, 1]$. The maximum is attained at the corners: when the household believes the shock is permanent ($\hat{\rho} = 1$) but it is fully transitory ($\rho = 0$), or vice versa. This bound is the sharpest test the model delivers. It does not require knowledge of $\hat{\rho}$: regardless of what the household believes, the rational forecast-error coefficient is bounded by ω.

A.3 Diagnostic Forecast

Under diagnostic expectations (Bordalo, Gennaioli, and Shleifer), the household distorts its likelihood by overweighting states made more representative by the recent signal. In the Gaussian case, this amounts to an additive distortion with parameter $\theta > 0$:

$$\mathbb{E}_t^\theta[\pi_{p,\,t\to t+1}] = \bar{\pi} + \omega\,\hat{\rho}\,s_{pt} + \theta\,\omega\,s_{pt}.$$

The diagnostic distortion is proportional to the signal s_{pt} and to the CPI weight ω. The parameter θ governs the degree of overweighting: $\theta = 0$ reduces to the rational case, and $\theta > 0$ generates over-reaction in the direction of the signal.

The diagnostic forecast error is

$$\begin{aligned} \mathrm{FE}^{\theta}_{pt} &= \omega(\rho - \hat{\rho})\, s_{pt} - \theta\, \omega\, s_{pt} + \eta_{pt} \\ &= \omega(\rho - \hat{\rho} - \theta)\, s_{pt} + \eta_{pt}. \end{aligned} \tag{11}$$

The regression coefficient is $\beta_3^{\theta} = \omega(\rho - \hat{\rho} - \theta)$. Since $\theta > 0$:

$$|\beta_3^{\theta}| > |\beta_3^{R}| \quad \text{for all } \rho, \hat{\rho} \in [0, 1].$$

The gap $|\beta_3^{\theta}| - |\beta_3^{R}| = \omega\, \theta$ identifies the diagnostic parameter scaled by the CPI weight.

A.4 Calibration

I estimate ρ from a pooled AR(1) regression on monthly province-level meat CPI log changes with province fixed effects. The estimating equation is

$$\Delta \log(\mathrm{MeatCPI}_{pm}) = \alpha_p + \rho\, \Delta \log(\mathrm{MeatCPI}_{p,m-1}) + \varepsilon_{pm},$$

with standard errors clustered at the province level. The sample covers January 2016 through December 2025 for 31 provinces, yielding approximately 3,689 monthly observations. The estimated persistence is $\hat{\rho} = 0.480$.

Given $\hat{\rho}$ from the data and $\omega = 0.05$, the rational benchmark is $|\beta_3^{R}| \leq \omega = 0.05$. The estimated $\hat{\beta}_3 = -6.067$ yields an implied diagnostic parameter of $\hat{\theta} = (|\hat{\beta}_3| - \omega)/\omega \approx (6.07 - 0.05)/0.05 = 120$. While the ordinal expectation scale inflates this number, the qualitative finding is robust: $\hat{\theta} \gg 0$.

A.5 Comparative Statics

The gap between the diagnostic and rational forecast errors is $|\beta_3^{\theta}| - |\beta_3^{R}| = \omega\, \theta$. This expression shows how the observable wedge depends on the model parameters.

The wedge is increasing in ω: a higher CPI weight on meat amplifies the overreaction because each unit of the diagnostic distortion translates into a larger revision of the expected headline inflation rate. In a country where meat carries a CPI weight of 0.10 rather than 0.05, the same θ would produce a wedge twice as large. This has implications for external validity: the overreaction documented here should be larger in economies with high food CPI weights and smaller in economies where food is a minor CPI component.

The wedge is increasing in θ: a more diagnostic household produces a larger forecast error for a given shock. The θ parameter is treated as a constant in the baseline model, but it could vary across households (for example, with education or financial literacy) or across signal types (for example, more salient signals could generate a higher effective θ). The null heterogeneity results in Table 7 suggest that θ does not vary strongly with observable demographics in this sample, though this does not rule out heterogeneity along unobserved dimensions such as news consumption or market experience.

The wedge does not depend on ρ or $\hat{\rho}$. The persistence parameters affect both the rational and diagnostic coefficients, but their effects cancel in the difference. This is a useful property: the identification of θ does not require the econometrician to know or estimate the household's belief about persistence. In practice, the calibration exercise in Section 6.4 uses ρ to compute the rational bound separately, but the excess beyond the bound identifies $\omega\,\theta$ regardless of ρ.

A.6 Connection to the Empirical Specification

The model maps directly to the regression equations in Section 5.2. The meat shock s_{pt} corresponds to the $\text{MeatShock}_{p,\,t-1\to t}$ variable constructed from NBS data. The forecast error FE_{pt} corresponds to $\pi^{\text{headline}}_{p,\,t\to t+1} - \tilde{\mu}_{ipt}$ in equation (7).

The model predicts that a regression of the forecast error on the meat shock should yield a coefficient β_3 that is bounded by ω under rationality. The empirical equation (7) includes fixed effects $\delta_{r(p)\times t}$ that are not present in the model. In the model, $\bar{\pi}$ is a constant; in the data, the fixed effects absorb variation in $\bar{\pi}$ across regions and waves, as well as any region-wave-specific component of η_{pt}. The fixed effects therefore improve the precision of $\hat{\beta}_3$ without affecting its probability limit, provided that the meat shock is conditionally uncorrelated with the fixed-effect-absorbed component of η_{pt}.

The household controls X_{ipt} in the empirical specification have no counterpart in the model. They are included to absorb observable heterogeneity in expectation formation (e.g., education, income, urban status) that might be correlated with provincial meat-shock exposure. Their inclusion does not alter the theoretical prediction; it sharpens the empirical estimate by reducing residual variance.

One gap between model and data deserves note. The model assumes that the household observes a continuous signal s_{pt} and forms a continuous expectation. The CFPS expectation is ordinal: $\mu_{ipt} \in \{-1, 0, 1\}$. The scaling from ordinal to CPI units (the mapping $\{-1, 0, 1\} \to \{-2, 0, 2\}$) introduces an additional degree of freedom that the model does not pin down. Appendix E shows that the sign of $\hat{\beta}_3$ is invariant to the scaling choice, so the qualitative test of rationality is unaffected. But the point estimate of $\hat{\theta}$ depends on the scaling and should be interpreted with caution.

A.7 Limitations of the Model

The signal-extraction framework is deliberately parsimonious. Its purpose is to provide a quantitative benchmark for the rational forecast error, not to describe the full information environment of Chinese households. Several features are omitted.

First, the model assumes a single signal. In reality, households observe multiple price signals (meat, grain, vegetables, housing, services) and must aggregate them into an inflation forecast. The model's prediction—that the forecast-error coefficient on the meat shock is bounded by ω—holds in a multi-signal extension provided that the meat shock is orthogonal to other signals conditional on fixed effects. The fixed-effect structure in the empirical specification is designed to absorb common components across signals.

Second, the model assumes static expectations: the household forms a one-period-ahead forecast and does not revise it. In practice, households may update their beliefs between survey waves as new information arrives. The biennial CFPS structure does not allow observation of intra-wave updating. A dynamic extension with sequential signals would be appropriate for higher-frequency data but would add parameters that the current data cannot identify.

Third, the model does not endogenise attention. The diagnostic parameter θ is a constant that applies to the meat signal regardless of its size. A richer model in the spirit of rational inattention (Sims) would allow θ to depend on the signal-to-noise ratio or the cost of processing information. Such an extension could explain why overreaction is larger during the ASF episode (when meat-price signals were unusually large) than in normal times, but it requires additional structure and cross-sectional variation in attention costs that are not available in the current data.

Fourth, the model assumes homogeneous households. In the data, households differ in income, education, and geographic location. A heterogeneous-agent extension would allow θ or $\hat{\rho}$ to vary with household characteristics. The null heterogeneity results suggest that any such variation is small, but the ordinal expectation measure may lack the resolution to detect moderate heterogeneity in the diagnostic parameter.

B Data Construction

This appendix documents the construction of the analysis dataset from raw sources, the coding decisions made at each stage, and the sample attrition from the raw CFPS to the estimation sample.

B.1 Province-Level CPI Data

The National Bureau of Statistics publishes monthly consumer price sub-indices at the province level. I collect the meat and poultry sub-index, the grain sub-index, the eggs sub-index, and the headline CPI index for all 31 mainland provinces, municipalities, and autonomous regions from January 2016 to December 2025. The data are sourced from the NBS monthly statistical releases and cross-checked against the CEIC China Premium Database.

Each observation is reported as a *previous month* = 100 index: a value of 105 indicates that the price level in the reference month is 5 percent higher than in the *preceding* month. The NBS publishes CPI sub-indices in multiple bases (previous month = 100, same month of preceding year = 100, and fixed base); I use the month-on-month series throughout because it permits exact construction of cumulative price changes over arbitrary calendar windows without requiring a year-on-year-to-monthly inversion.

I convert each monthly observation to a log change, $\Delta \log(\mathrm{CPI}_{pm}) = \log(\mathrm{CPI}_{pm}/100)$.

Because the raw index uses the previous month as base, $\mathrm{CPI}_{pm}/100$ equals the gross monthly price change, and its logarithm is the exact month-on-month log growth rate. For values close to 100 the approximation is accurate to second order; for extreme values (such as meat CPI reaching 138 during the ASF peak) the log transformation compresses the upper tail relative to a linear percentage change.

The treatment variable for the meat shock is the cumulative sum of monthly log changes between the midpoints of consecutive CFPS fieldwork periods:

$$\mathrm{MeatShock}_{p,\,t-1\to t} = \sum_{m\in\mathcal{M}(t-1,t)} \Delta\log(\mathrm{MeatCPI}_{pm}).$$

The fieldwork midpoints are approximated as July of each wave year (2016, 2018, 2020, 2022), based on the typical CFPS survey timeline. The grain shock and eggs shock are constructed identically from their respective sub-indices. All three commodity shocks vary at the province-wave level and enter the regressions as right-hand-side variables.

B.2 CFPS Panel Construction

The China Family Panel Studies is a nationally representative longitudinal survey conducted by the Institute of Social Science Survey at Peking University. I merge the household-level data across waves 2016, 2018, 2020, and 2022. The key variables are the inflation expectation item (qp201 in most waves), household demographic characteristics, and the province of residence.

The inflation expectation variable is coded as $\mu_{it} \in \{-1, 0, 1\}$, where -1 indicates the household expects prices to fall, 0 indicates prices will stay the same, and 1 indicates prices will rise. The original CFPS coding uses numeric values (typically 1, 2, 3 or 1, 2, 5 depending on the wave), which I recode to the $\{-1, 0, 1\}$ ordinal scale. I verify the recoding against the CFPS codebook for each wave to ensure consistency.

Province-of-residence codes present a harmonization challenge. Across waves, the CFPS uses different variable names: `provcd` in some waves, `provcd10` in the 2010 baseline, `provcd12` in the 2012 follow-up, and variants in later waves. All province codes follow the GB/T 2260 national standard but differ in the number of trailing digits. I standardise all codes to the two-digit level (e.g., Beijing = 11, Shanghai = 31, Guangdong = 44) and merge on the standardised code. Provinces that changed administrative boundaries during the sample period (none did at the provincial level between 2016 and 2022) would require additional harmonization, but this issue does not arise.

The province code is used to match each household to province-level CPI data. The merge is many-to-one: all households within a province at a given wave receive the same province-wave meat shock. The CFPS covers 25 of the 31 mainland provinces in the public-use data; the remaining provinces are either not sampled or suppressed for confidentiality. I use all available provinces, which yields coverage of approximately 95 percent of the national population.

B.3 Forecast Error Construction

The forecast error is defined as

$$\mathrm{FE}_{ipt} = \pi^{\text{headline}}_{p,\,t\to t+1} - \tilde{\mu}_{ipt},$$

where $\pi^{\text{headline}}_{p,\,t\to t+1}$ is the annualised realised headline CPI inflation in province p over the next inter-wave period, and $\tilde{\mu}_{ipt}$ is the household's ordinal expectation scaled to CPI units.

The forward CPI is computed as follows. For each province and wave, I take the cumulative sum of monthly headline CPI log changes from the midpoint of wave t to the midpoint of wave $t+1$. This cumulative change is then annualised by dividing by the number of years between midpoints (approximately 2 for biennial waves). The annualisation ensures that the forecast error is expressed in annual percentage-point units, comparable across wave pairs of slightly different length.

For the scaling of the ordinal expectation, I map $\{-1, 0, 1\}$ to $\{-2, 0, 2\}$ percentage points in the baseline specification. This mapping reflects the assumption that a household reporting "prices will rise" expects approximately 2 percent annual inflation, a household reporting "stay the same" expects zero, and a household reporting "prices will fall" expects approximately 2 percent deflation. These values are chosen to be symmetric and to approximate the unconditional mean of headline CPI inflation during the sample period (which averages approximately 2 percent). Appendix E reports sensitivity to alternative scaling choices, including $\{-1, 0, 1\}$ (unit ordinal), $\{-4, 0, 4\}$ (wide), and

$\{-1, 0, 3\}$ (asymmetric).

The forecast-error sample requires that forward realised CPI be available. For the 2022 wave, the forward CPI requires data through mid-2024, which is available in the current data extract. For any wave whose forward CPI is not yet observed at the time of data construction, the observation is dropped from the forecast-error estimation but retained for the expectation equation (Eq 1), which does not use forward CPI.

B.4 Sample Attrition

The raw CFPS household panel contains approximately 130,000 household-wave observations across the four waves (2016, 2018, 2020, 2022). The estimation sample is substantially smaller due to several restrictions.

The first restriction is non-missing inflation expectations. Approximately 15 to 20 percent of households do not respond to the inflation expectation question (item qp201) in a given wave, either because the question was not administered (routing differences across waves) or because the respondent refused or answered "don't know." This restriction reduces the sample to approximately 105,000 observations.

The second restriction is a valid province code that matches the CPI data. Households in provinces not covered by the CFPS public-use data, or with missing province codes, are dropped. This restriction has a modest effect, reducing the sample to approximately 100,000 observations.

The third restriction is non-missing meat-shock values. Because the NBS meat CPI data begin in January 2016, the earliest wave for which the meat shock can be constructed is 2016 (using the 2014–2016 backward link). Households surveyed before 2016, if any remain after wave selection, are dropped. After this restriction, the full estimation sample for the expectation equation (Eq 1) contains 90,133 household-wave observations.

The fourth restriction applies only to the forecast-error equation (Eq 3): forward realised headline CPI must be available. For wave t, this requires CPI data through the midpoint of wave $t + 1$. At the time of data construction, forward CPI is available for the 2016, 2018, and 2020 waves (using data through mid-2022, mid-2024, and projected mid-2024 respectively). The 2022 wave lacks a complete forward realisation at the time of the most recent data pull, reducing the forecast-error sample to 69,533 observations.

The fifth restriction applies to specifications with demographic controls. The age, education, and urban-status variables are drawn from the CFPS household and individual questionnaires. The urban indicator is missing for a substantial fraction of observations in the 2018, 2020, and 2022 waves due to changes in the questionnaire routing. When all three demographic controls are required simultaneously, the sample shrinks to approximately 22,121 observations. The main results are reported with and without demographic controls to show that the forecast-error finding is not driven by the sample composition change.

B.5 Persistence Estimation

The AR(1) persistence of meat-price shocks is estimated from the monthly province-level data. The estimating equation is

$$\Delta \log(\text{MeatCPI}_{pm}) = \alpha_p + \rho \, \Delta \log(\text{MeatCPI}_{p,m-1}) + \varepsilon_{pm},$$

estimated by OLS with province fixed effects and standard errors clustered at the province level. The sample covers January 2016 to December 2025 across 31 provinces, yielding 3,689 monthly observations after dropping the initial month for each province. The estimated coefficient is $\hat{\rho} = 0.480$ (s.e. $= 0.010$, $p < 0.001$).

The AR(1) specification is chosen for parsimony: it provides a single persistence parameter that maps directly into the signal-extraction model. Higher-order specifications (AR(2) or AR(12) to capture seasonal patterns) would allow richer dynamics but would complicate the calibration without changing the qualitative conclusion, because the rational bound depends on the total persistence over the forecast horizon, not on the lag structure. The province fixed effects absorb persistent differences in the level of meat-price inflation across provinces, ensuring that $\hat{\rho}$ captures within-province temporal persistence rather than cross-province level differences.

C Wild Cluster Bootstrap

C.1 Methodology

Asymptotic cluster-robust standard errors can be unreliable with fewer than approximately 50 clusters (Cameron, Gelbach, and Miller; MacKinnon and Webb). With 31 province clusters, I supplement conventional inference with the wild cluster bootstrap. This appendix describes the methodology in detail and reports the full results.

The wild cluster bootstrap constructs the null distribution of the test statistic by resampling residuals at the cluster level. Two versions of the procedure exist: the wild cluster unrestricted (WCU) and the wild cluster restricted (WCR). In the WCU version, the bootstrap samples are constructed from residuals of the unrestricted model (which includes the coefficient of interest). In the WCR version, the bootstrap samples are constructed from residuals of the restricted model, which imposes the null hypothesis $H_0 \colon \beta = 0$.

I use the WCR version throughout. MacKinnon and Webb show that the restricted bootstrap has superior size control when the number of clusters is small. The intuition is that the restricted residuals are computed under the null, so the bootstrap distribution is centred at the correct location under H_0. The unrestricted version uses residuals computed under the alternative, which can introduce a bias in the bootstrap distribution when the number of clusters is small and the test statistic is far from zero. With 31 clusters and a t-statistic exceeding 5 for the forecast-error coefficient, this bias is potentially nontrivial, making the restricted version the more conservative choice.

I use the six-point weight distribution proposed by Webb, which is designed for settings with few clusters. The standard wild bootstrap draws weights from a two-point distribution—either Rademacher $\{-1, +1\}$ or Mammen $\{-(1-\sqrt{5})/2, (1+\sqrt{5})/2\}$—with equal probability. With $G = 31$ clusters, the Rademacher distribution produces $2^{31} \approx 2.15 \times 10^9$ distinct bootstrap samples. While this is a large number, the effective coverage of the null distribution can be uneven because each bootstrap draw assigns the same weight to all observations within a cluster. The six-point distribution provides a richer set of weights $\{w_1, \ldots, w_6\}$ with $6^{31} \approx 1.3 \times 10^{24}$ distinct samples, which yields a finer approximation to the true null distribution. Webb show in simulations with 10–30 clusters that the six-point distribution produces rejection rates closer to the nominal size than either the Rademacher or Mammen distributions.

Each bootstrap replication proceeds as follows:

1. Estimate the restricted model (imposing $H_0\colon \beta = 0$) and obtain residuals $\hat{\varepsilon}^r_{ig}$ for each observation i in cluster g.
2. Draw a six-point random weight w_g independently for each of the 31 province clusters. The weight takes one of six values, each with probability 1/6.
3. Construct the bootstrap error as $\varepsilon^*_{ig} = w_g \cdot \hat{\varepsilon}^r_{ig}$. All observations within a cluster receive the same weight, preserving the within-cluster correlation structure.
4. Construct the bootstrap dependent variable: $y^*_{ig} = \hat{y}^r_{ig} + \varepsilon^*_{ig}$, where $\hat{y}^r_{ig}$ is the fitted value from the restricted model.
5. Re-estimate the unrestricted model on (y^*_{ig}, X_{ig}) and compute the t-statistic t^*_b on the coefficient of interest.
6. Repeat steps 2–5 for $B = 9{,}999$ replications.
7. The bootstrap p-value is the fraction of replications for which $|t^*_b| \geq |t_{\text{obs}}|$, where t_{obs} is the test statistic from the original data.

The choice of $B = 9{,}999$ is standard in the literature and ensures that the bootstrap p-value has a precision of approximately ± 0.01 at the 5 percent level. With $B = 9{,}999$ and a true rejection rate of 0.05, the Monte Carlo standard error of the estimated p-value is $\sqrt{0.05 \times 0.95/10{,}000} \approx 0.002$.

C.2 Results

Table A1 reports the bootstrap results for the two main coefficients (expectation and forecast error) under both fixed-effect structures.

Table A1: Wild Cluster Bootstrap Results

Equation	FE structure	$\hat{\beta}$	SE	Asymptotic p	Bootstrap p	N
Eq 1 (Expectations)	Prov + Wave	0.090	0.053	0.100	0.088	90,133
Eq 1 (Expectations)	Region×Wave	0.242	0.174	0.174	0.237	90,133
Eq 3 (Forecast error)	Prov + Wave	-5.622	1.182	< 0.001	< 0.001	69,533
Eq 3 (Forecast error)	Region×Wave	-6.067	1.145	< 0.001	< 0.001	69,533

Webb six-point distribution with 9,999 replications. Restricted bootstrap (WCR) imposing $H_0\colon \beta = 0$. p-values are two-sided.

The forecast-error coefficient $\hat{\beta}_3$ is robust to the bootstrap correction under both fixed-effect structures. The bootstrap p-values for Eq 3 are below 0.001 in both cases, matching the asymptotic p-values. The 95-percent bootstrap confidence intervals for $\hat{\beta}_3$ exclude zero by a wide margin under both specifications.

The expectation coefficient $\hat{\beta}_1$ is marginal. Under province-plus-wave FE, the bootstrap p-value of 0.088 is slightly below the conventional 10 percent threshold, while the asymptotic p-value of 0.100 is at the boundary. Under region-by-wave FE, both the asymptotic (0.174) and bootstrap (0.237) p-values are above 10 percent. The bootstrap does not reverse any asymptotic conclusion but does widen the p-value modestly for Eq 1 under region-by-wave FE, reflecting the conservative nature of the WCR procedure with few clusters.

C.3 Power Considerations

The wild cluster bootstrap corrects the size of the test (the probability of rejecting a true null) but does not improve its power (the probability of rejecting a false null). With 31 clusters, the effective degrees of freedom for cluster-level variation are limited, and power against small to moderate effect sizes is low.

This power limitation is relevant for interpreting the imprecise expectation coefficient. The identifying variation in $\hat{\beta}_1$ operates at the province-wave level: with 31 provinces and 4 waves, the maximum number of province-wave cells is 124, but the effective number is reduced by the fixed-effect structure. Under region-by-wave FE with approximately 24 region-wave cells, the residual degrees of freedom for the meat-shock coefficient are approximately $124 - 24 = 100$ at the cell level, but cluster-robust inference treats the 31 provinces as the unit, yielding an effective sample size of 31 for the purpose of the t-distribution. Power calculations for a two-sided test at the 5 percent level with 30 degrees of freedom suggest that detecting an effect of the size estimated for $\hat{\beta}_1$ (0.242 with s.e. 0.174, implying a t-ratio of 1.4) requires approximately 50–60 clusters to achieve 80 percent power. The current design with 31 clusters is underpowered for the expectation coefficient.

The forecast-error coefficient does not face this power problem. The t-ratio for $\hat{\beta}_3$ exceeds 5 under both FE structures, which is well above any critical value for 30 degrees of freedom. The bootstrap confirms that the apparent power is not an artifact of size distortion: the p-values remain below 0.001 after the correction.

The asymmetry in power between Eq 1 and Eq 3 reflects both the measurement properties of the dependent variables and the structure of the identifying variation. The ordinal expectation in Eq 1 takes three values $(-1, 0, 1)$, which limits the effective variation in the dependent variable. The forecast error in Eq 3 combines the ordinal expectation with a continuous CPI measure, producing a dependent variable with substantially more dispersion. The greater dispersion in the dependent variable of Eq 3 translates to a higher signal-to-noise ratio for the coefficient of interest and, consequently, a larger t-statistic for a given effect size.

A further consideration is that the wild cluster bootstrap is designed for inference, not for estimation. The point estimates $\hat{\beta}_1$ and $\hat{\beta}_3$ are OLS estimates and are unbiased under the standard assumptions. The bootstrap corrects only the distribution of the test statistic. The fact that the bootstrap widens the p-value for $\hat{\beta}_1$ under region-by-wave FE (from 0.174 to 0.237) does not imply that the point estimate is biased; it implies that the asymptotic standard error slightly understates the true sampling variability of the t-ratio with 31 clusters.

These power considerations motivate the emphasis on the forecast-error equation as the primary test. The forecast-error coefficient provides the most powerful test of overreaction because it combines the expectation and CPI channels into a single dependent variable with high dispersion, and because the sign of $\hat{\beta}_3$ alone is sufficient to reject the rational benchmark without requiring a precise estimate of the expectation response.

D Alternative Fixed-Effect Structures

This appendix provides a detailed comparison of the two fixed-effect structures used in the main analysis, explains the identifying variation under each, and reports the key coefficients in a unified format.

D.1 Fixed-Effect Specifications

The main text reports results under two fixed-effect structures: region-by-wave (the preferred specification) and province plus wave (the robustness check). Each structure absorbs a different component of the variation in outcomes and treatment, leaving a different residual source of identification.

Under province-plus-wave additive two-way fixed effects, the specification includes a set of province dummies α_p and a set of wave dummies δ_t, entered additively:

$$y_{ipt} = \beta\, s_{pt} + \gamma' X_{ipt} + \alpha_p + \delta_t + \varepsilon_{ipt}.$$

Province fixed effects absorb all time-invariant province characteristics: average income levels, baseline food culture, persistent differences in CPI basket composition, quality of local statistical reporting, and any other province-specific factor that does not change across waves. Wave fixed effects absorb all nationwide shocks common to a given survey round: national monetary policy stance, global commodity-price movements, national media coverage of inflation, and survey-round effects (e.g., changes in questionnaire wording or fieldwork procedures).

Under this structure, the identifying variation is the province-specific deviation of the meat shock from the national wave mean, after removing province means. Formally, $\tilde{s}_{pt} = s_{pt} - \bar{s}_{p\cdot} - \bar{s}_{\cdot t} + \bar{s}_{\cdot\cdot}$, the standard within-estimator residual. This variation captures the fact that some provinces experienced larger meat-price increases in a given wave relative to both their own average and the national average for that wave.

Under region-by-wave interacted fixed effects, the specification groups the 31 provincial units into macro regions and includes region-by-wave interactions:

$$y_{ipt} = \beta\, s_{pt} + \gamma' X_{ipt} + \delta_{r(p)\times t} + \varepsilon_{ipt},$$

where $r(p)$ maps province p to one of six macro regions (East, Central, West, Northeast, and in some classifications further splits). With four waves and six regions, there are approximately 24 region-by-wave cells.

Region-by-wave fixed effects absorb any shock common to all provinces within a region at a given wave: regional business cycles, regional monetary policy transmission differentials, regionally correlated weather or disease shocks, and regional media markets. This is a stronger absorber than the additive two-way structure because it allows the national wave effect to differ across regions. For example, if the ASF shock hit northeastern provinces earlier and harder than southern provinces, the two-way structure would fail to absorb this regional timing difference, while the interacted structure would absorb it.

The identifying variation under region-by-wave FE is the within-region cross-province deviation of the meat shock: $\tilde{s}_{pt} = s_{pt} - \bar{s}_{r(p),t}$, the meat shock for province p in wave t minus the average meat shock for provinces in the same region and wave. This is a narrower source of variation than the two-way residual and produces fewer effective degrees of freedom.

D.2 What Each Structure Absorbs

Table A2 summarises the sources of variation absorbed by each FE structure and the residual identifying variation.

Table A2: Comparison of Fixed-Effect Structures

Source of variation	Prov + Wave	Region×Wave
Province time-invariant level	Absorbed	Absorbed[a]
National wave effect	Absorbed	Absorbed
Region-specific wave effect	*Not absorbed*	Absorbed
Province-specific time trend	*Not absorbed*	*Not absorbed*
Within-region province×wave	Identifying	Identifying

[a] Province time-invariant levels are subsumed by region-by-wave FE only to the extent that region membership captures the cross-province level variation. Province-specific deviations from the regional mean are not absorbed.

The key difference is the treatment of region-specific wave effects. Under additive two-way FE, a demand boom in the eastern coastal provinces during the 2020 wave would remain in the residual and could confound the meat-shock coefficient if the demand boom also affected meat prices. Under region-by-wave FE, this variation is absorbed. The cost is a reduction in identifying variation: instead of using all 31 provinces relative to the national mean, the interacted structure uses provinces only relative to their regional peers.

D.3 Coefficient Comparison Across Structures

Table A3 reports the key coefficients from Equations 1 and 3 under both FE structures, along with bootstrap p-values.

Table A3: Key Coefficients Under Alternative Fixed-Effect Structures

Equation	FE structure	$\hat{\beta}$	SE	Bootstrap p	N
Panel A: Expectation equation (Eq 1)					
	Prov + Wave	0.090	0.053	0.088	90,133
	Region×Wave	0.242	0.174	0.237	90,133
Panel B: Forecast-error equation (Eq 3)					
	Prov + Wave	−5.622	1.182	< 0.001	69,533
	Region×Wave	−6.067	1.145	< 0.001	69,533

Wild cluster bootstrap (WCR) with Webb six-point distribution and 9,999 replications.
Standard errors clustered at the province level.
p-values are two-sided.

The forecast-error coefficient is stable across FE structures. Moving from province-plus-wave to region-by-wave, the point estimate shifts from −5.622 to −6.067, a change of 7.9 percent. Both estimates are precisely estimated with $p < 0.001$ under both asymptotic and bootstrap inference. The modest increase in magnitude under region-by-wave FE suggests that absorbing regional trends removes a small positive confound (i.e., a regional factor that slightly attenuates the negative forecast-error relationship), but the adjustment is minor relative to the overall coefficient.

D.4 Interpretation of Coefficient Stability

The stability of $\hat{\beta}_3$ across FE structures is informative about the nature of the identifying variation. If the meat-shock coefficient were driven by a regional confound—for example, if northeastern provinces simultaneously experienced large meat-price increases and negative economic shocks that independently lowered inflation—then moving from additive to interacted FE should attenuate the coefficient substantially. The observed stability indicates that the within-region cross-province variation produces the same estimate as the pooled variation, which is consistent with the meat shock operating through a household-level salience channel rather than through a region-level macroeconomic confound.

The expectation coefficient behaves differently. The point estimate increases from

0.090 under province-plus-wave to 0.242 under region-by-wave, but the standard error more than triples (from 0.053 to 0.174). The increase in the point estimate reflects the removal of a negative confound at the regional level: under additive FE, the estimate is pulled toward zero by regional factors that covary negatively with both the meat shock and the expectation. The large increase in the standard error reflects the mechanical reduction in identifying variation: the region-by-wave cells absorb a substantial share of the province-level shock, leaving little residual variation for identification.

This precision loss is expected and does not undermine the main finding. The forecast-error test (Eq 3) is the primary identification strategy precisely because it does not require a precise estimate of the expectation coefficient. The sign of $\hat{\beta}_3$ provides a direct test of overreaction that is robust to the FE structure, while the magnitude of $\hat{\beta}_1$ provides mechanism evidence whose precision varies with the specificity of the FE absorption.

D.5 Why Not Province-by-Wave Fixed Effects?

The most saturated option would be province-by-wave fixed effects. With 31 provinces and 4 waves, this would create 124 province-wave cells, leaving zero degrees of freedom for the province-wave-level meat shock. The meat shock varies at the province-wave level, so it would be perfectly collinear with province-by-wave FE and cannot be estimated. The region-by-wave structure is the most demanding FE specification that preserves identification of the treatment variable.

E Expectation Scaling Sensitivity

E.1 The Scaling Problem

The CFPS inflation expectation question is ordinal. Respondents report whether they expect prices to rise, stay the same, or fall. The raw response is coded as $\mu_{it} \in \{-1, 0, 1\}$. To construct the forecast error $\text{FE}_{ipt} = \pi^{\text{headline}}_{p,\, t\to t+1} - \tilde{\mu}_{ipt}$, I must convert the ordinal expectation into units comparable with the realised CPI (which is measured in annual percentage points). This conversion requires a scaling function $\tilde{\mu} = f(\mu)$ that maps ordinal codes to CPI units.

The scaling is inherently underdetermined. The CFPS question asks for the direction of expected price changes, not for a point forecast. A respondent who reports "prices will rise" could mean 1 percent, 5 percent, or 20 percent. No information in the survey resolves this ambiguity. The scaling choice therefore affects the level and variance of the forecast error and, consequently, the magnitude (though not the sign) of the regression coefficient $\hat{\beta}_3$.

This section evaluates the sensitivity of the main results to the scaling choice by considering four alternative mappings and providing economic reasoning for each.

E.2 Scaling Alternatives

I consider the following four scaling functions.

Baseline: $\{-2, 0, 2\}$. The baseline maps "fall" to -2 percentage points, "same" to 0, and "rise" to $+2$ percentage points. The rationale is that unconditional headline CPI inflation in China during the sample period (2016–2022) averaged approximately 2 percent per year. A respondent who expects "prices will rise" is interpreted as expecting roughly average inflation. A respondent who expects "prices will fall" is interpreted as expecting deflation of comparable magnitude. The symmetry of the mapping reflects the symmetric phrasing of the survey question.

Alternative 1: $\{-1, 0, 1\}$ (unit ordinal). The simplest scaling uses the ordinal codes directly. The forecast error becomes realised CPI in percentage points minus the ordinal code (-1, 0, or 1). This scaling implicitly assumes that "prices will rise" corresponds to 1 percent expected inflation. Given that average realised inflation is approximately 2 percent, this mapping understates the expected inflation of optimistic respondents and compresses the forecast-error distribution.

Under this scaling, the dependent variable has a narrower range. The forecast error for a respondent in a province with 2 percent realised inflation who reported "rise" is $2 - 1 = 1$ percentage point, compared with $2 - 2 = 0$ under the baseline. The coefficient $\hat{\beta}_3$ will be larger in magnitude under the unit scaling because the dependent variable attributes less of the realised CPI to the expectation component, making the forecast error larger on average for "rise" respondents.

Alternative 2: $\{-4, 0, 4\}$ (wide scaling). This scaling assumes that "prices will rise" implies roughly 4 percent expected inflation. The rationale is that during high-inflation episodes (such as the 2019–2020 food-price spike), households reporting "rise" may have had in mind inflation rates well above the unconditional mean. If respondents anchor on recent salient price increases rather than on average CPI, a wider scaling may better approximate the implied expectation.

Under this scaling, the forecast error for a "rise" respondent in a 2-percent-inflation province is $2 - 4 = -2$ percentage points. The wider scaling mechanically increases the fraction of negative forecast errors among "rise" respondents, reducing the variance of the forecast error conditional on the shock and potentially attenuating $\hat{\beta}_3$.

Alternative 3: $\{-1, 0, 3\}$ (asymmetric). This scaling allows the "rise" response to carry a different magnitude than the "fall" response. The rationale is that China experienced positive average inflation during the sample period, so "prices will rise" may be the default expectation (consistent with the fact that approximately 60 to 70 percent of respondents choose this category in most waves). If "rise" is the typical response, the associated expected inflation may be higher than the expected deflation associated with "fall." The asymmetric mapping assigns $+3$ to "rise" and -1 to "fall," reflecting the possibility that respondents who expect prices to rise anticipate moderate inflation while those who expect prices to fall anticipate only mild deflation.

E.3 Results Under Alternative Scalings

Table A4 reports the forecast-error coefficient $\hat{\beta}_3$ under each scaling for the preferred region-by-wave FE specification.

Table A4: Forecast-Error Coefficient Under Alternative Expectation Scalings

Scaling	$\tilde{\mu}$ map	$\hat{\beta}_3$	SE	p-value	N
Baseline	$\{-2,\ 0,\ 2\}$	−6.067	1.145	< 0.001	69,533
Unit ordinal	$\{-1,\ 0,\ 1\}$	−6.309	1.319	< 0.001	69,533
Wide	$\{-4,\ 0,\ 4\}$	−5.583	0.978	< 0.001	69,533
Asymmetric	$\{-1,\ 0,\ 3\}$	−5.825	1.067	< 0.001	69,533

Region-by-wave fixed effects. Standard errors clustered at the province level.
All specifications include the same set of controls as the preferred specification in Table 2.

The sign of $\hat{\beta}_3$ is negative and statistically significant at $p < 0.001$ under all four scalings. The point estimate ranges from −5.583 (wide scaling) to −6.309 (unit ordinal scaling), a spread of approximately 12 percent around the baseline. The variation in magnitude follows the logic outlined above. A wider positive scaling (attributing more expected inflation to "rise" respondents) mechanically reduces the forecast error and compresses $\hat{\beta}_3$ toward zero, while a narrower scaling inflates the forecast error and increases $|\hat{\beta}_3|$.

E.4 Why the Sign Is Invariant

The invariance of the sign to scaling has a simple algebraic explanation. The forecast error is $\mathrm{FE}_{ipt} = \pi_{p,t\to t+1} - f(\mu_{it})$. The meat shock enters the regression through its covariance with the forecast error. The realised CPI $\pi_{p,t\to t+1}$ does not depend on the scaling (it is an external variable), so the only channel through which scaling affects $\hat{\beta}_3$ is through its effect on $f(\mu_{it})$.

Since the ordinal expectation μ_{it} is positively correlated with the meat shock (households in high-shock provinces are more likely to report "rise"), the scaled expectation $f(\mu_{it})$ is also positively correlated with the shock for any scaling that is monotonically increasing in μ (i.e., any scaling where "rise" maps to a higher value than "fall"). The forecast error $\pi - f(\mu)$ therefore has a negative partial correlation with the shock because the expectation component overshoots the CPI component. This negative partial correlation produces a negative $\hat{\beta}_3$ regardless of the specific values assigned to each ordinal category, as long as the ordering is preserved.

The magnitude of $\hat{\beta}_3$ changes with the scaling because the scaling affects the variance of $f(\mu)$ and its covariance with the shock. A wider scaling increases the variance of the expectation component, which reduces the forecast error for "rise" respondents (making it less negative) and increases it for "fall" respondents (making it more positive). The net effect on $\hat{\beta}_3$ depends on the relative magnitudes of these adjustments, but the sign is determined by the ordering of the categories, which is invariant to monotone transformations.

E.5 Which Scaling Is Most Defensible?

No scaling can be verified from the data because the survey does not elicit a point forecast. However, three considerations favour the baseline $\{-2, 0, 2\}$ mapping.

First, the baseline is centred on the unconditional mean of realised headline CPI during the sample period (approximately 2 percent per year). This makes the forecast error approximately mean-zero for respondents who correctly predict the direction of inflation, which is a desirable property for interpreting the constant in the regression.

Second, the symmetric mapping avoids imposing an asymmetry on the expectation distribution that the survey question does not support. The CFPS question is phrased symmetrically ("will prices rise, stay the same, or fall?"), so there is no a priori reason to assign different magnitudes to the "rise" and "fall" categories. The asymmetric scaling $\{-1, 0, 3\}$ is motivated by the empirical distribution of responses (most respondents choose "rise"), but this distribution could reflect either the level of expected inflation or a cognitive default toward the modal response. Without additional information, the symmetric mapping is more conservative.

Third, the baseline scaling produces a coefficient magnitude (-6.067) that is intermediate in the range across scalings. This means that the baseline is neither the most nor the least conservative interpretation of the overreaction. The qualitative comparison with the rational benchmark ($|\beta_3^R| \leq 0.05$) is unaffected: the estimated coefficient exceeds the bound by a factor of at least 112 (under the wide scaling) and at most 126 (under the unit ordinal scaling). The two-orders-of-magnitude excess is invariant to any reasonable scaling choice.

E.6 Implications for the Implied Diagnostic Parameter

The implied diagnostic parameter $\hat{\theta} = |\hat{\beta}_3|/\omega$ depends on both the scaling and the assumed CPI weight ω. Under the baseline scaling and $\omega = 0.05$, $\hat{\theta} \approx 121$. Under the unit ordinal scaling, $\hat{\theta} \approx 126$; under the wide scaling, $\hat{\theta} \approx 112$; under the asymmetric scaling, $\hat{\theta} \approx 117$.

These values are all large relative to estimates of θ in the professional-forecast literature, where Bordalo et al. report values on the order of 0.5 to 1.5. The inflation of

$\hat{\theta}$ in this setting is expected: the ordinal expectation measure is a coarse proxy for the continuous belief that the diagnostic model assumes. The three-category variable compresses the rich heterogeneity in household expectations into three bins, amplifying the apparent diagnostic parameter. A continuous expectation measure (e.g., a point forecast in percentage terms) would produce a lower $\hat{\theta}$ because it would capture gradations within the "rise" category that the ordinal variable conflates.

The scaling exercise confirms that the ordinal nature of the expectation measure introduces quantitative imprecision in the implied θ but does not affect the qualitative conclusion. The estimated forecast error exceeds the rational benchmark by a factor of at least 50 under every scaling considered, and the sign of $\hat{\beta}_3$ is invariant to the choice of mapping.

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